

Bölüm 4

Ortografik Çizimler



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İçerik

- Ortografik Çizimler
 - Tüm Basamaklar
 - Görünüş seçimi için öneriler
- Görünüş seçimi için örnekler
- İzdüşüm Sistemleri
- 1.Bölge Sistemine Göre Standart Görünüşler
- Öncelikli Yardımcı Görünüşler



Ortografik Çizimler

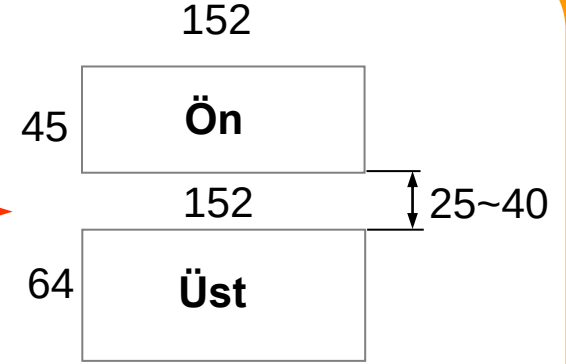
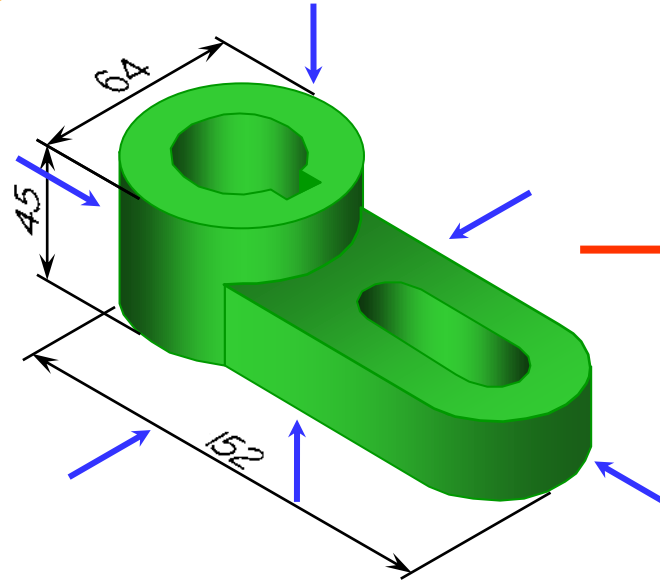
Adım Adım Çizim

1. Gerekli Görünüřleri Seç

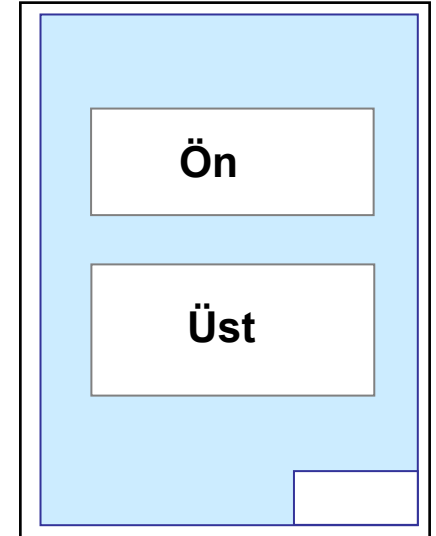
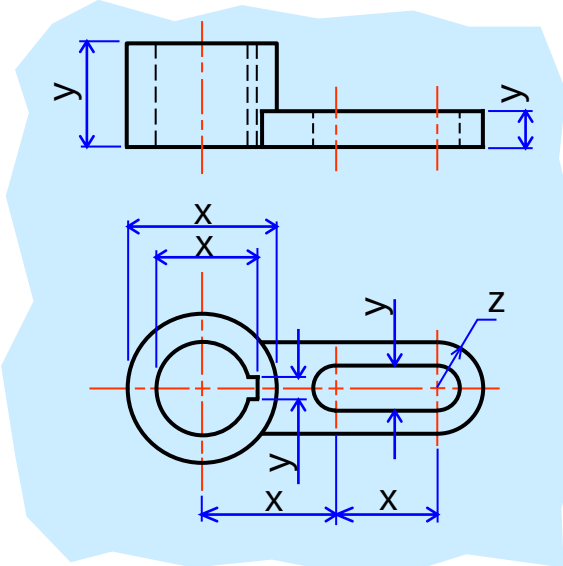
2. Seçilen görüntüleri çizim kağıdına yerleştir

3. Seçilen görüntüleri tamamla

4. Ölçü ve açıklama bilgilerini tamamla



Ölçek seç
(1:1)



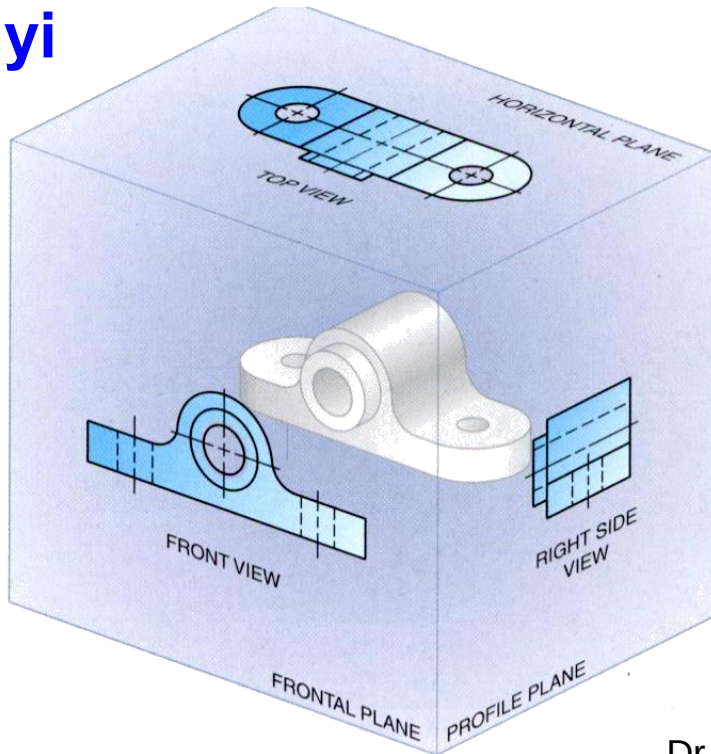
Görünüş Seçme Prosedürleri

1. Cismi, Cam kutuya (3.Bölge Sitemi) en iyi pozizyonda yerleştir.
2. Ön görünüş için en uygun yüzeyi seç
3. Ön görünüşe uygun diğer görüşleri seç.

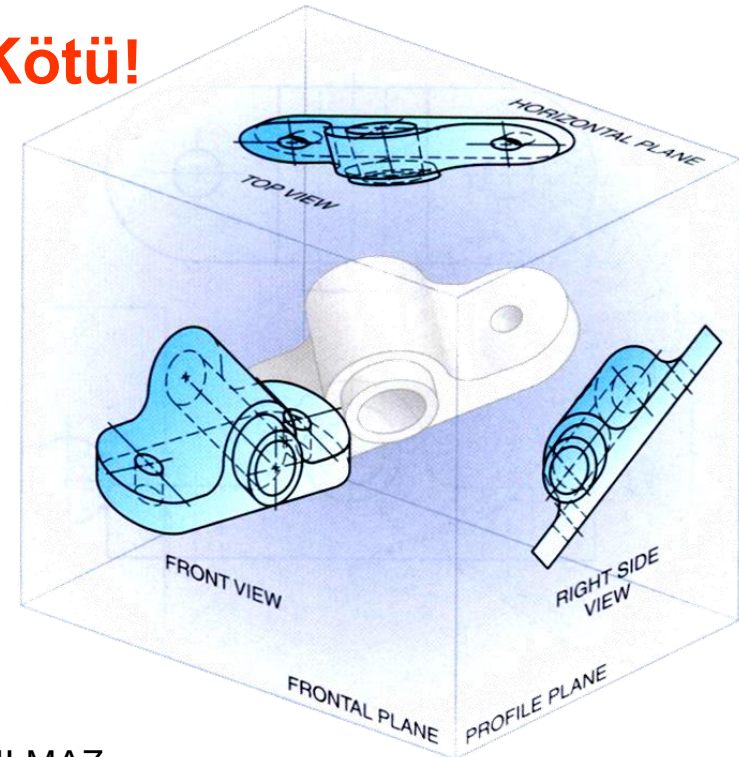
Öneriler : Cismin Yerleştirilmesi

- 1.Cismi **normal pozisyonunda** yerleştirin
 - 2.Orthographic görüntüler, mümkün olduğunca cismin **gerçek boyut ve şeklini** göstermelidir.
-

İyi



Kötü!

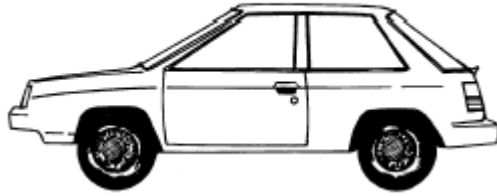


ÖN GÖRÜNÜŞ SEÇİMİ

1. Cismin **en uzun** kenarına sahip yüzey **ön görünüş** olarak seçilir.

İlk seçim

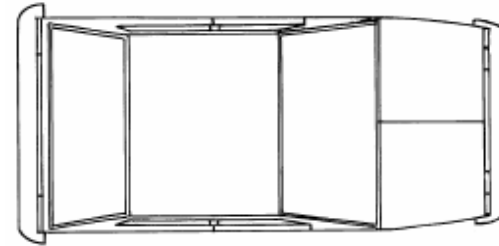
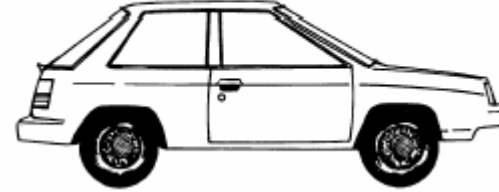
Uygun değil



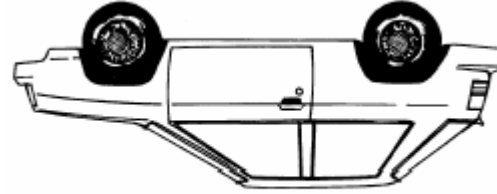
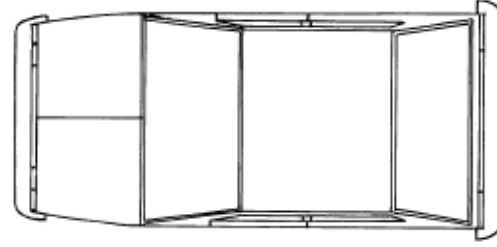
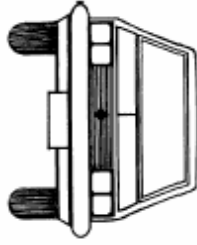
Daha fazla alan kaplar

İkinci seçim

Uygun

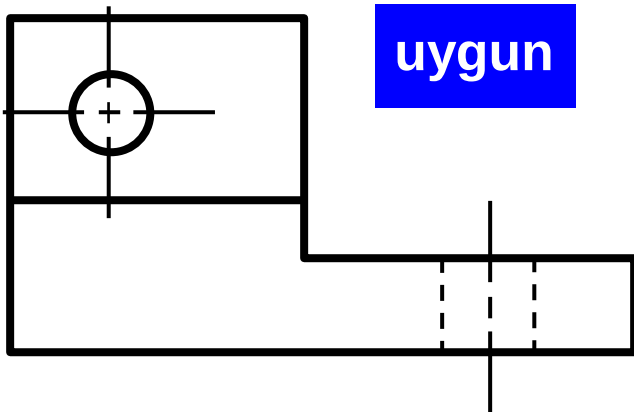
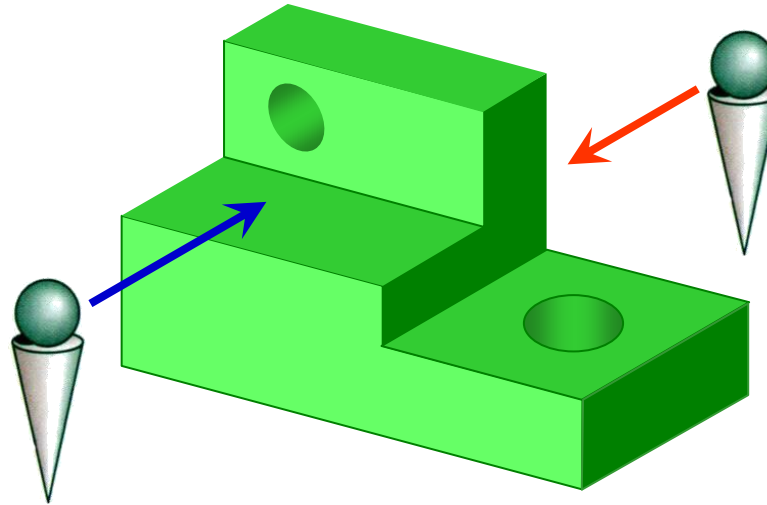


2. Diğer görüntüler seçilen ön görünüşe uygun olarak çizilir.

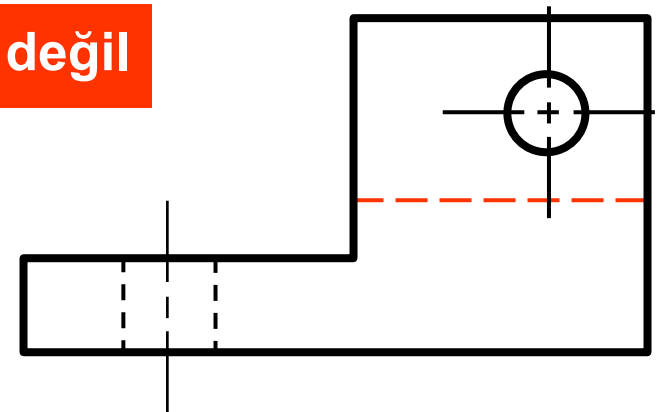


Uygun değil

3. Görünmez çizgilerin **en az** olduğu yön **ön** görünüş olarak seçilmeli

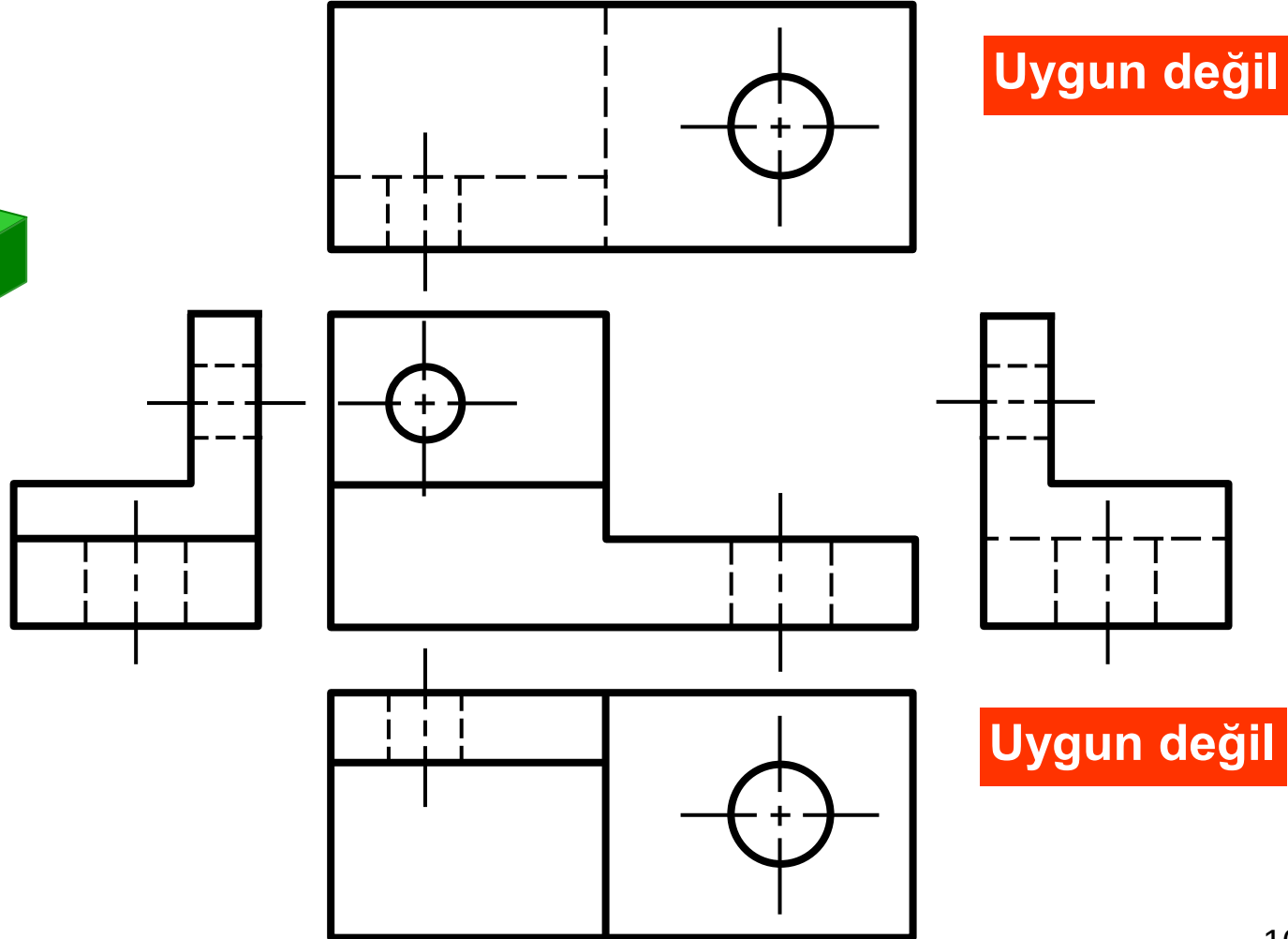
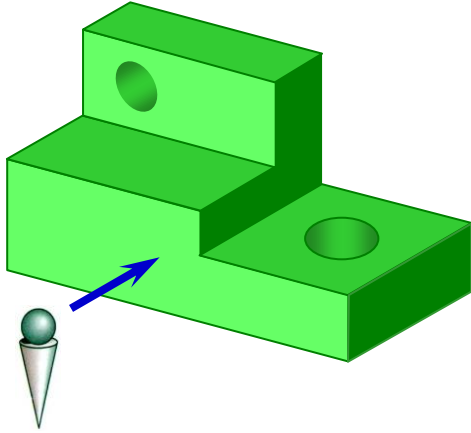


Uygun değil

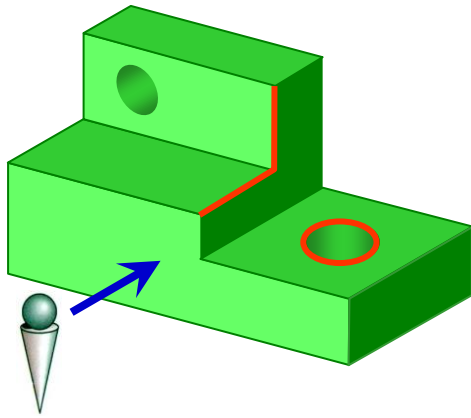


Öneriler: Komşu Görünüşlerin Seçimi

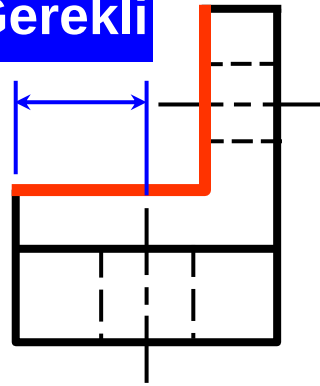
1. Görünmez çizgilerin en az olduğu görünüş seçilmeli.



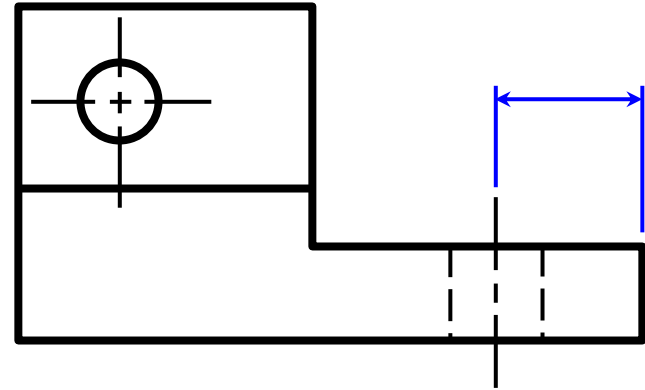
2. Cismin maksimum özelliği, minimum görünüş sayısı ile verilmeli.



Gerekli

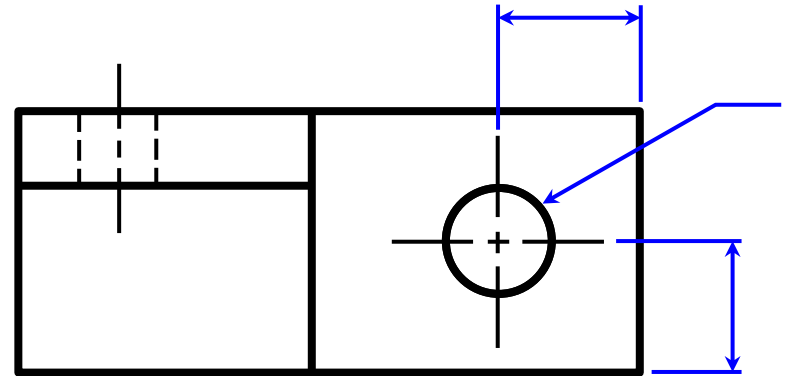


Delikle ilgili bilgiler ayrı bir görünüşte verilir.

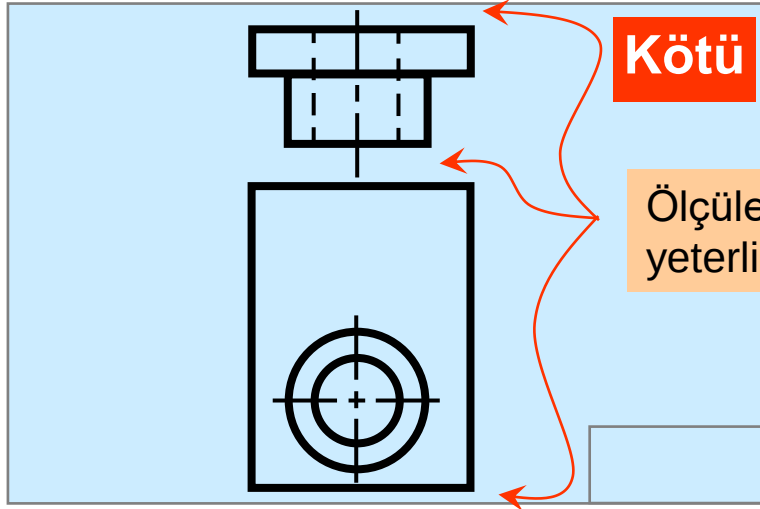
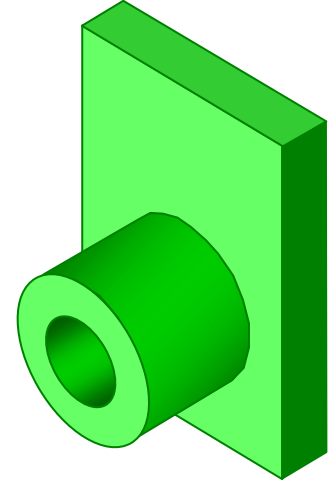


Gerekli

Bütün bilgiler tek görünüşte verilir.

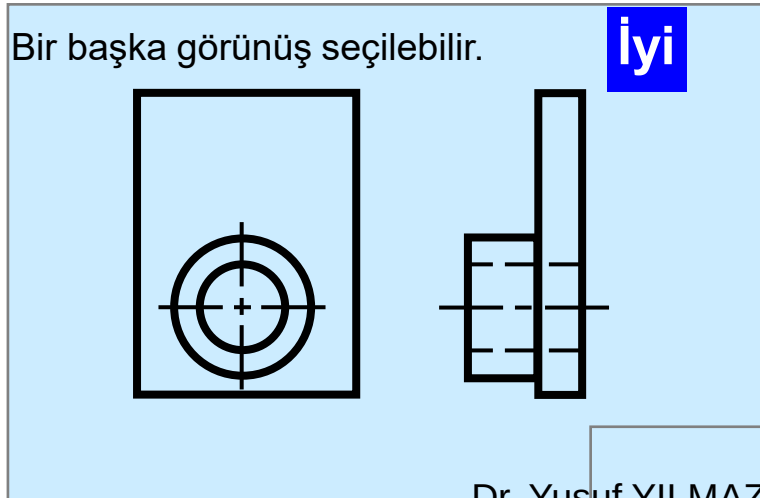


3. Görünüşler, çizim kağıdında minimum yer alacak şekilde yerleştirilir



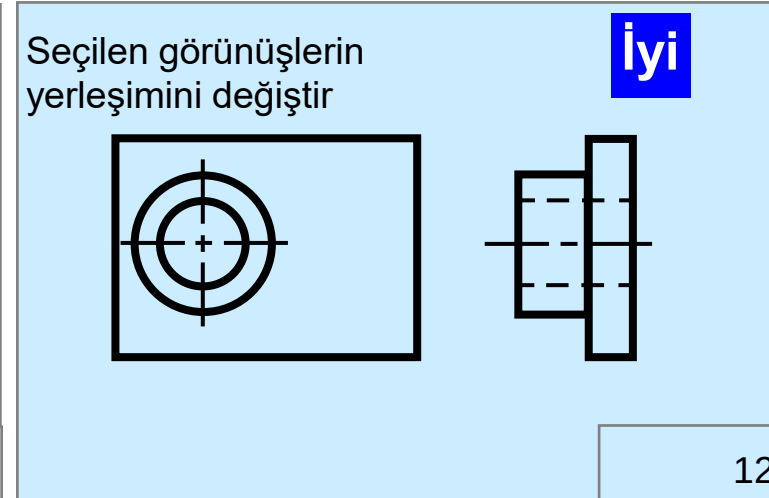
Kötü

Ölçülendirme için yeterli yer yok.



Bir başka görünüş seçilebilir.

İyi



Seçilen görünüşlerin yerleşimini değiştir

İyi

Özetle;

- Üç basamakta görünüş seçimi



- Pratikte, Teknik ressam çizime başlamadan önce bütün seçenekleri gözden geçirmeli.

Kağıda yerleşimi,
görüntü seçimi vs.



Görünüş Seçimi İle İlgili Örnekler

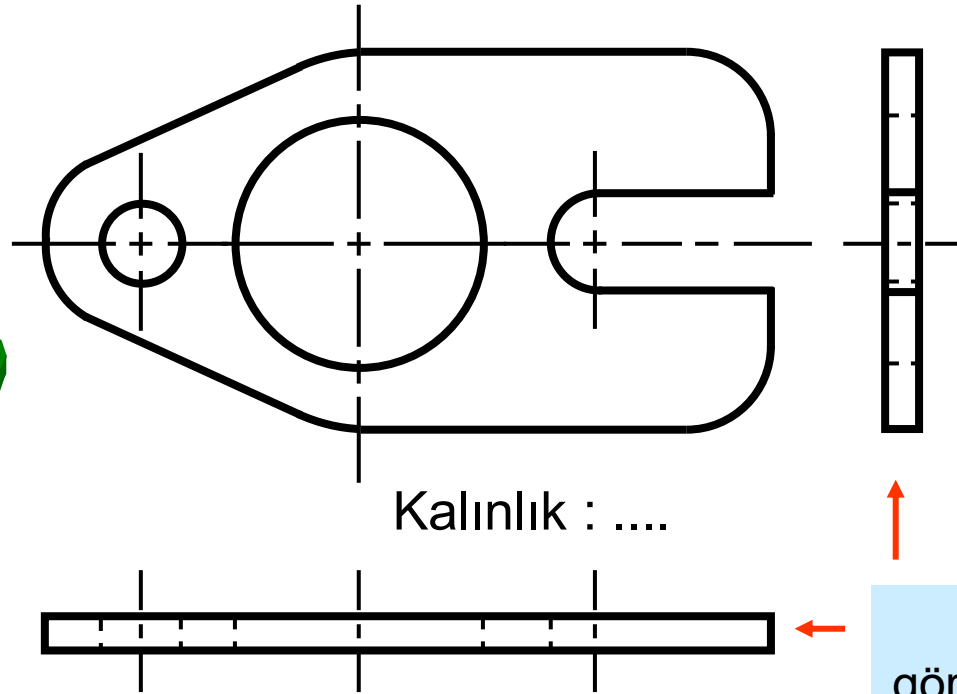
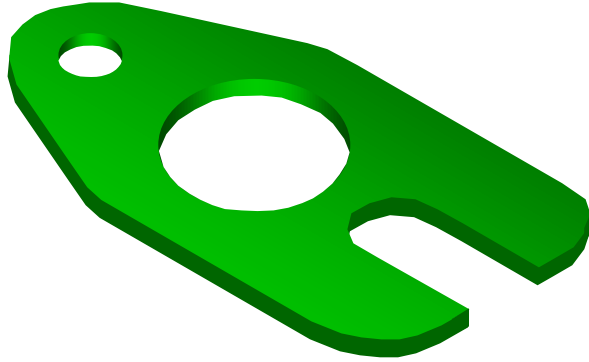
Notlar

- Genellikle, cismi tam olarak ifade edebilecek üç görünüş seçilmeli
- Bazı özel durumlarda gerekli görünüş sayısı üçten az yada fazla olabilir.

Tek Görünüşün Yeterli Olduğu Cisimler

- Sac metal ve Conta gibi sabit kesitli ince parçalar.
- Silindirik şekilli parçalar

Örnek



Kalınlık :

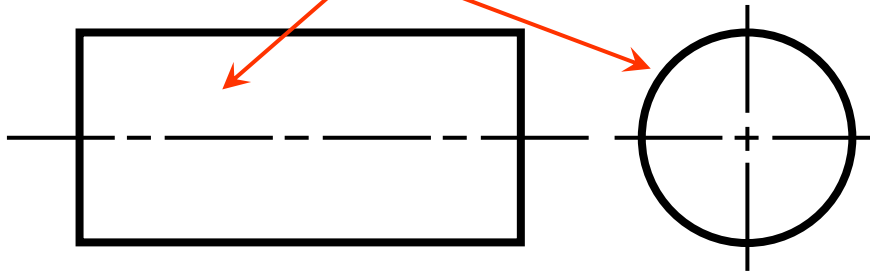
Yardımcı
görünüşler parça
kalınlığı için
gerekli !

■ Sac metal ve conta gibi sabit kesitli ince parçalar.

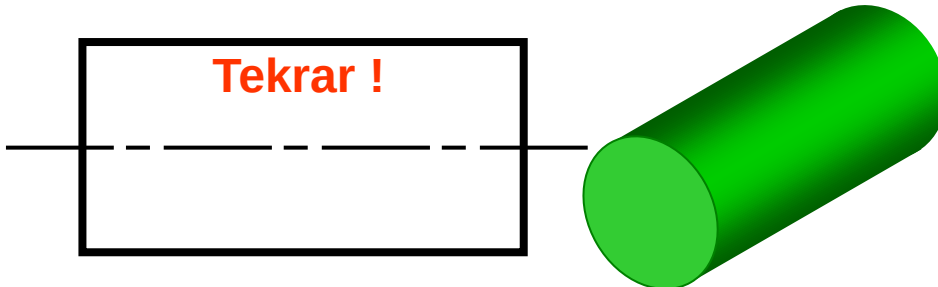
■ Silindirik şekilli parçalar

Örnek

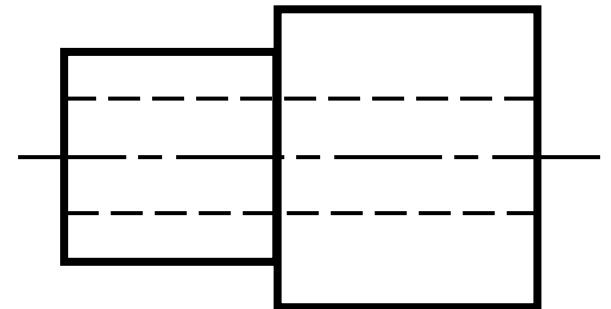
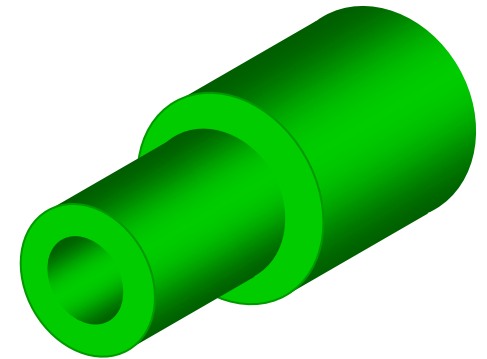
Silindirden çıkarılabilir.



Tekrar !



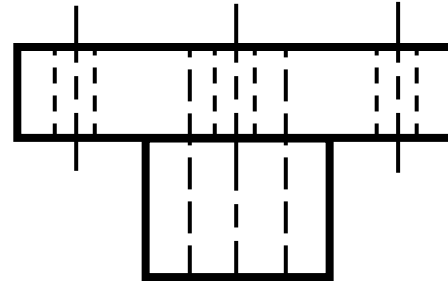
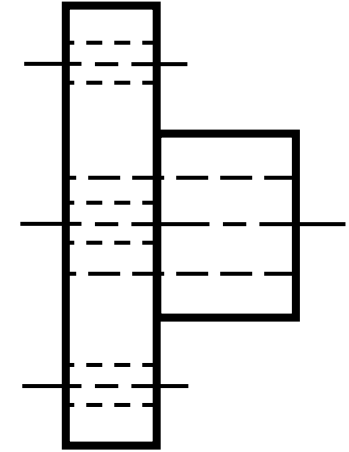
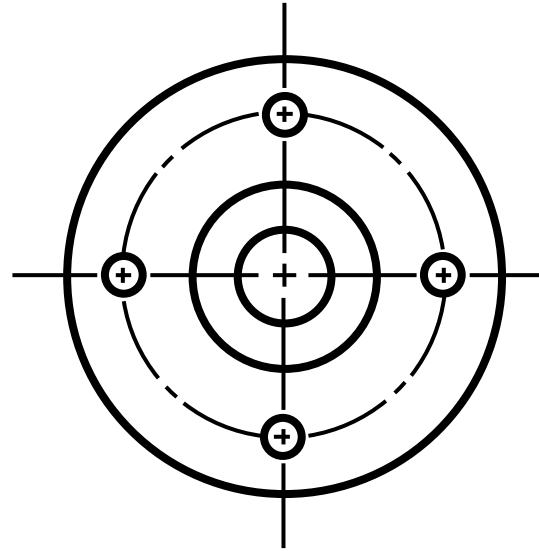
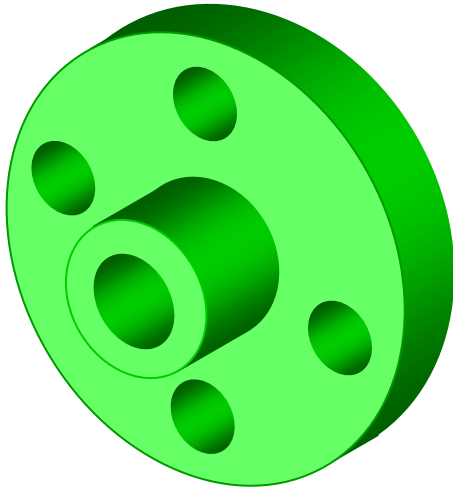
Örnek



İki Görünüşün Yeterli Olduğu Cisimler

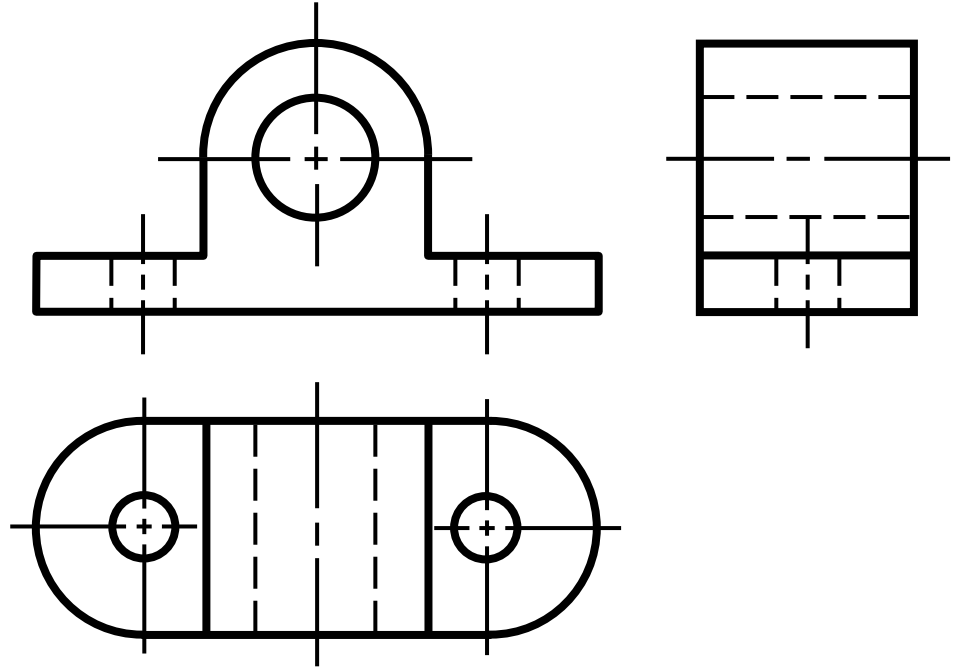
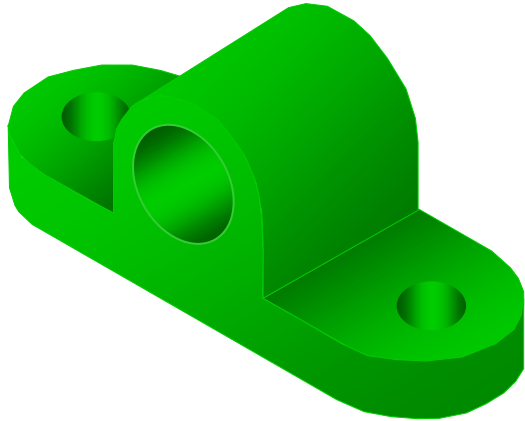
- Yardımcı görünüşlerin her ikisi de aynı bilgi ve şekle sahip ise.

Örnek

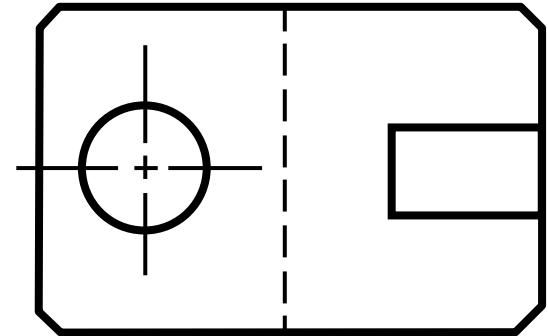
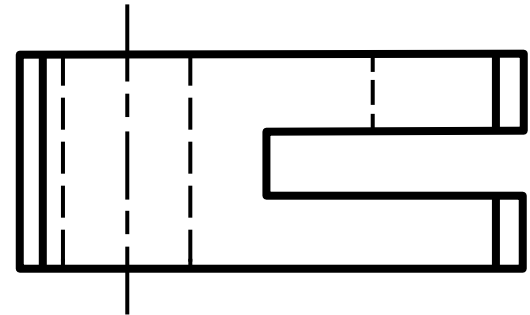
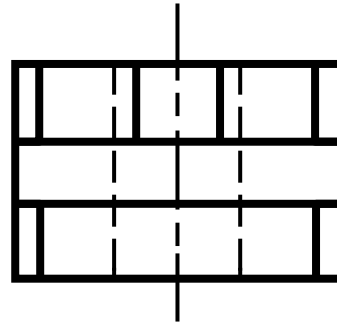
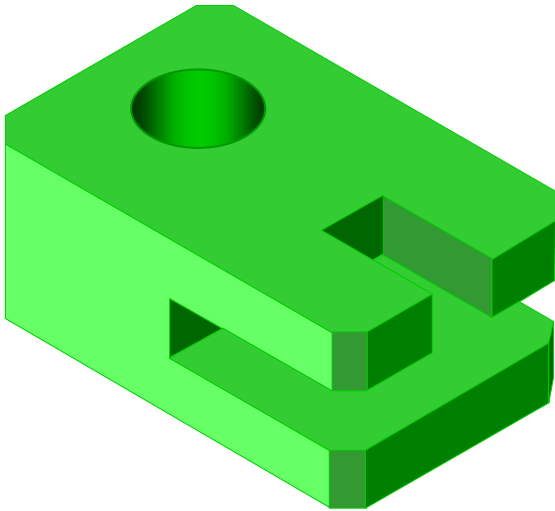


Tekrar!

Örnek1



Örnek 2





İzdüşümü Sistemleri

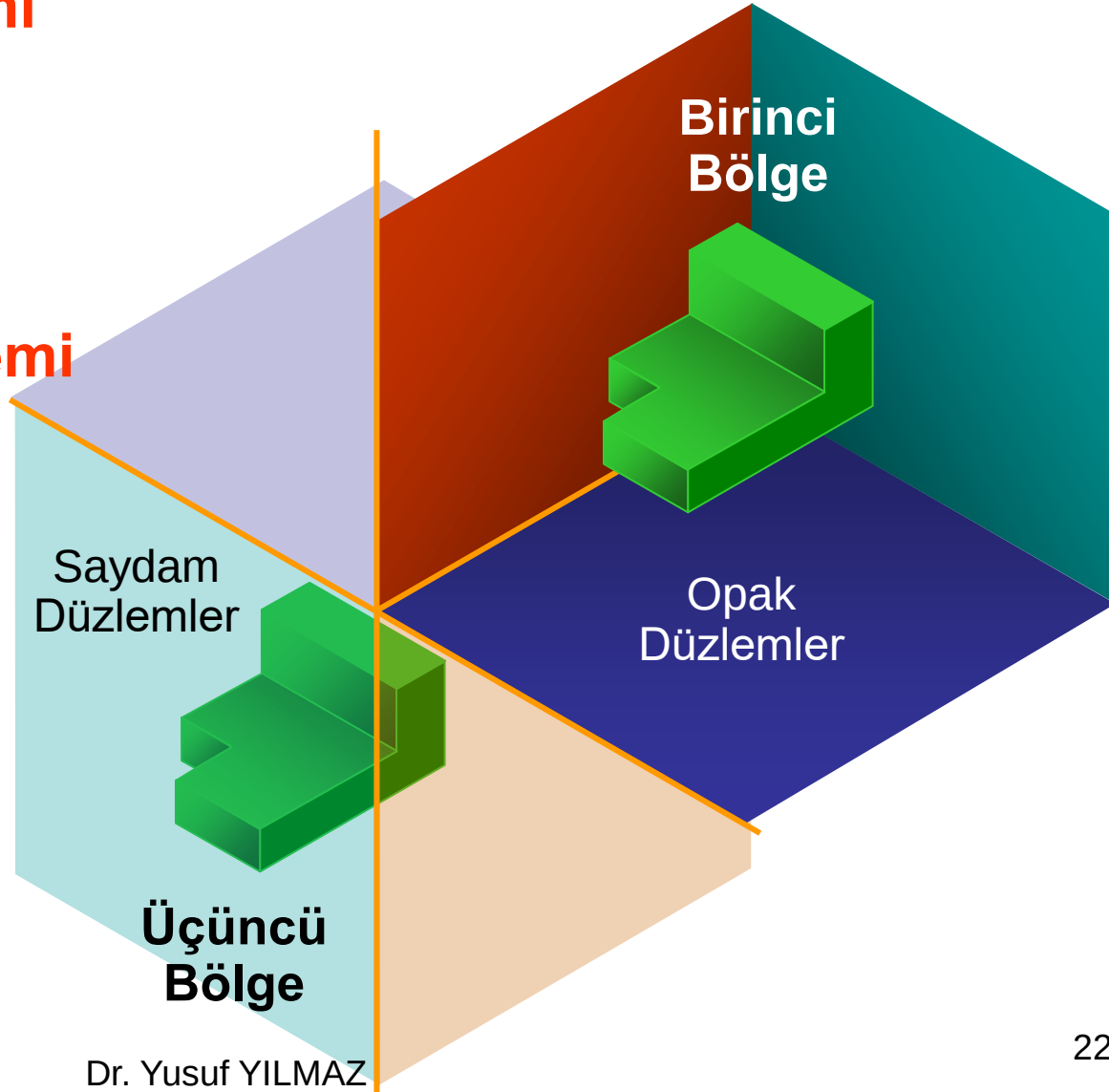
İzdüşümü Sistemleri

1. Birinci Bölge Sistemi

- Avrupa Ülkeleri
- ISO standartları

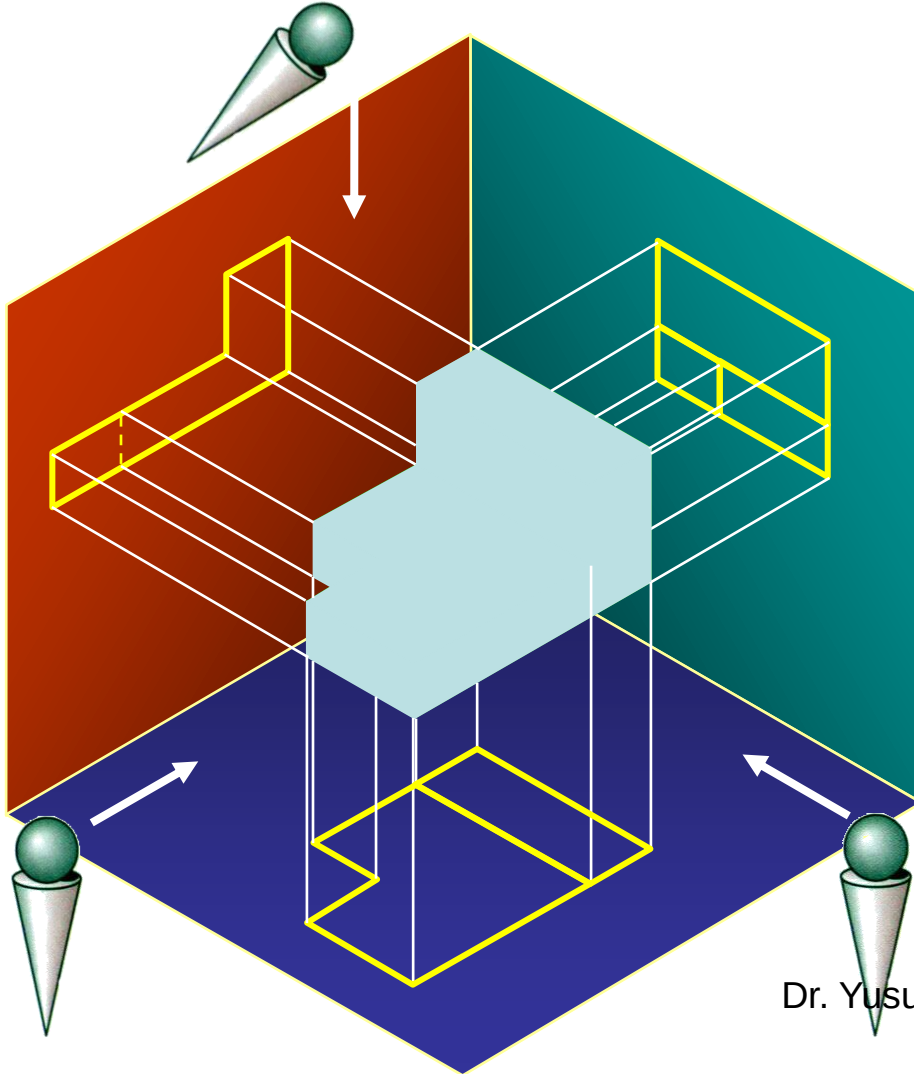
2. Üçüncü Bölge Sistemi

- Kanada,
- ABD,
- Japonya,
- Tayland

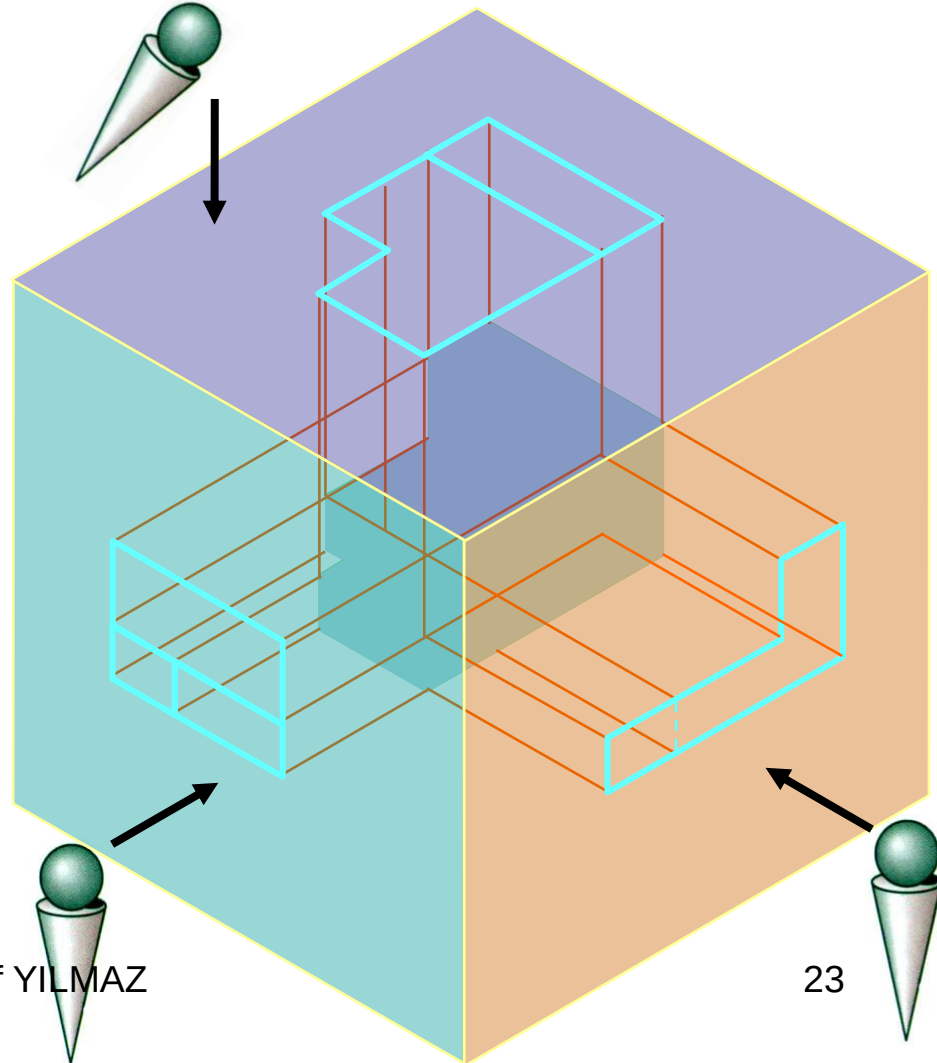


Ortografik Görünüşler

1.Bölge Sistemi
(Opak Düzlemler)

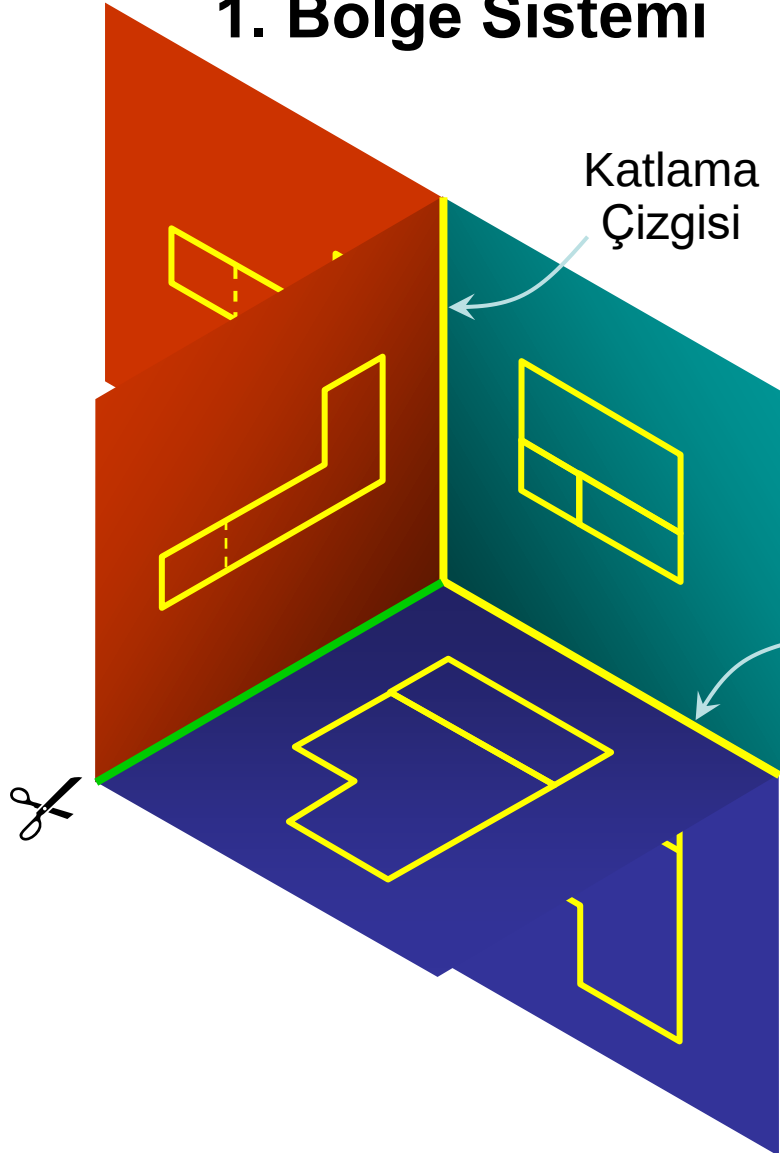


3.Bölge Sistemi
(Saydam Düzlemler / Cam Kutu)

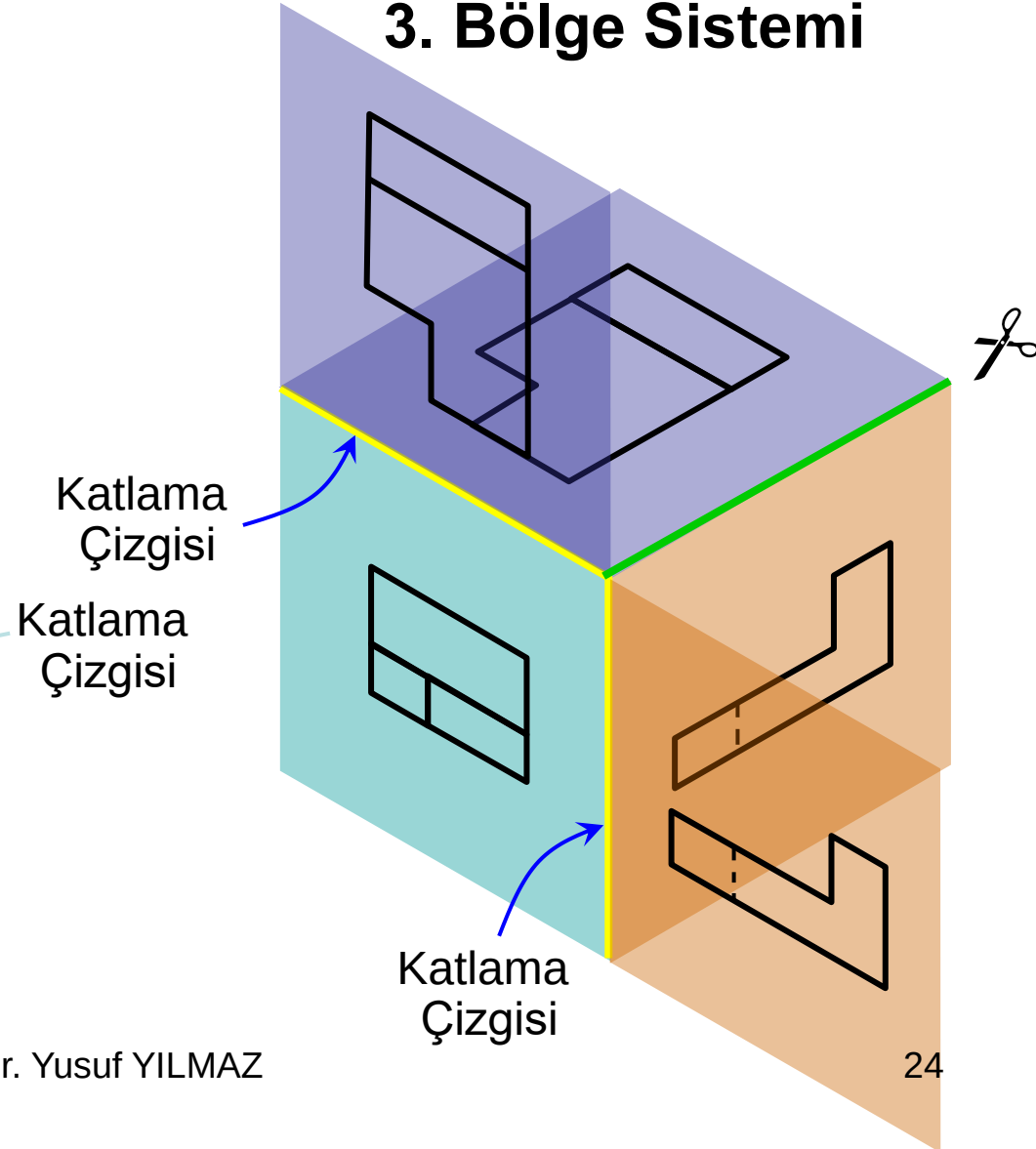


Ortografik Görünüşler

1. Bölge Sistemi

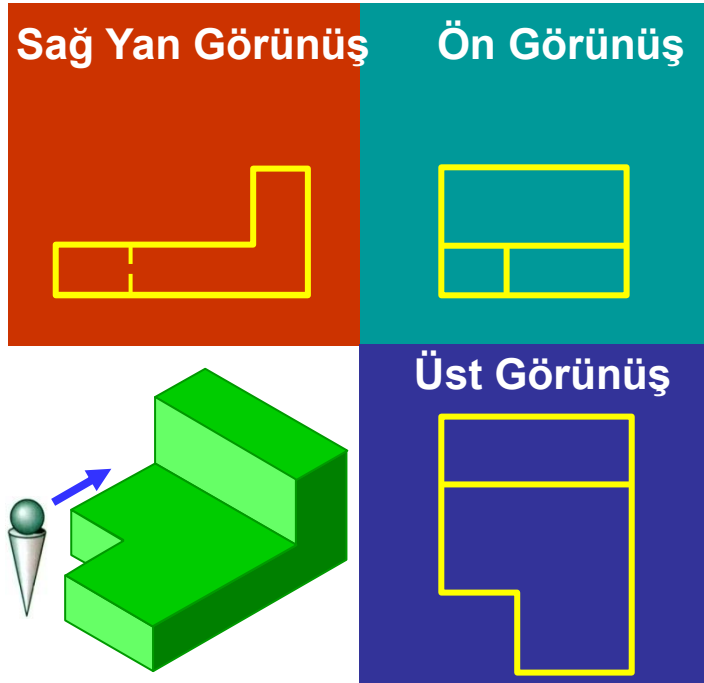


3. Bölge Sistemi

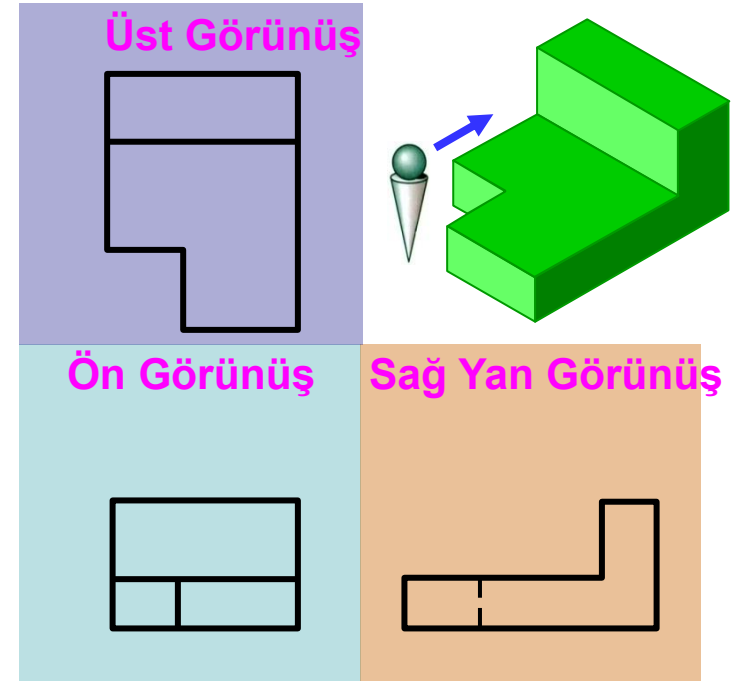


Görünüşlerin Düzenlenmesi

1. Bölge Sistemi

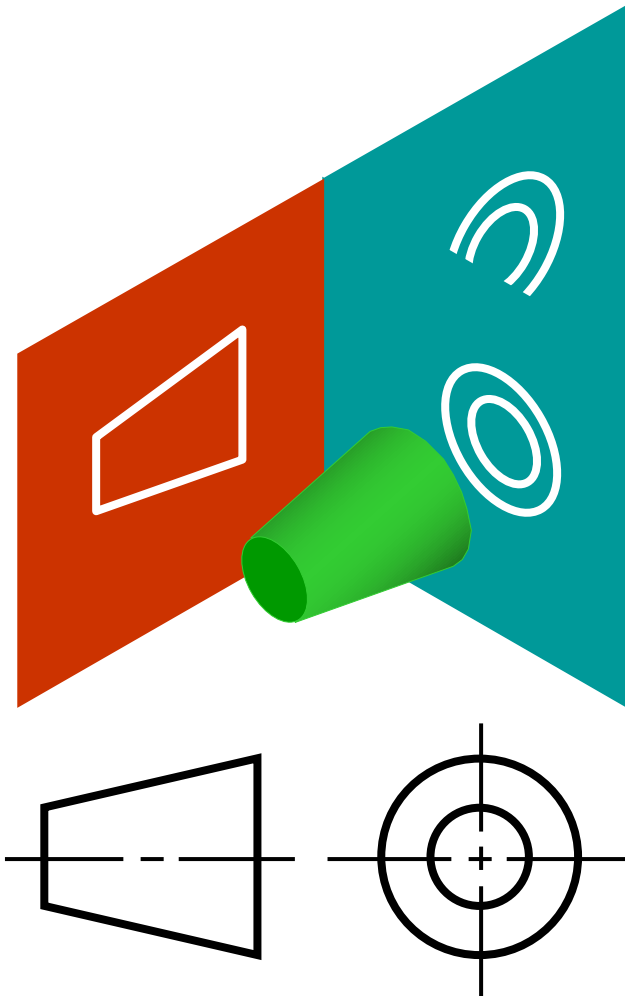


3. Bölge Sistemi

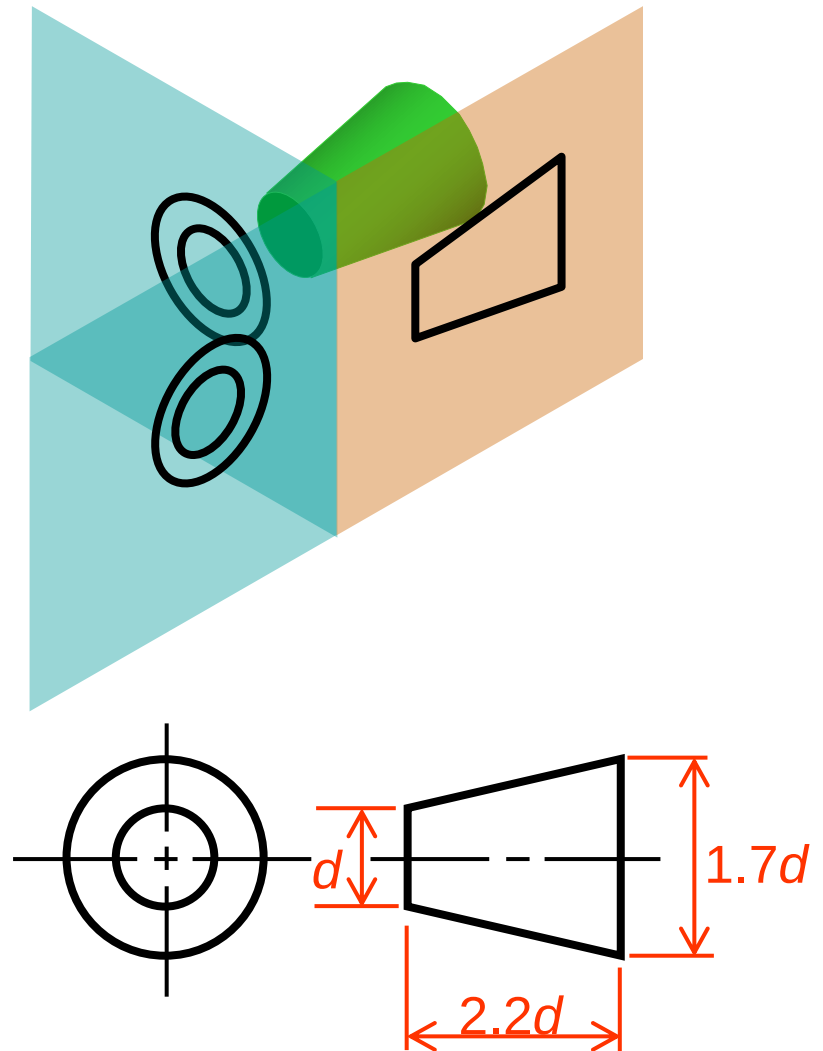


İzdüşümü Bölge Sembolleri

1. Bölge Sistemi

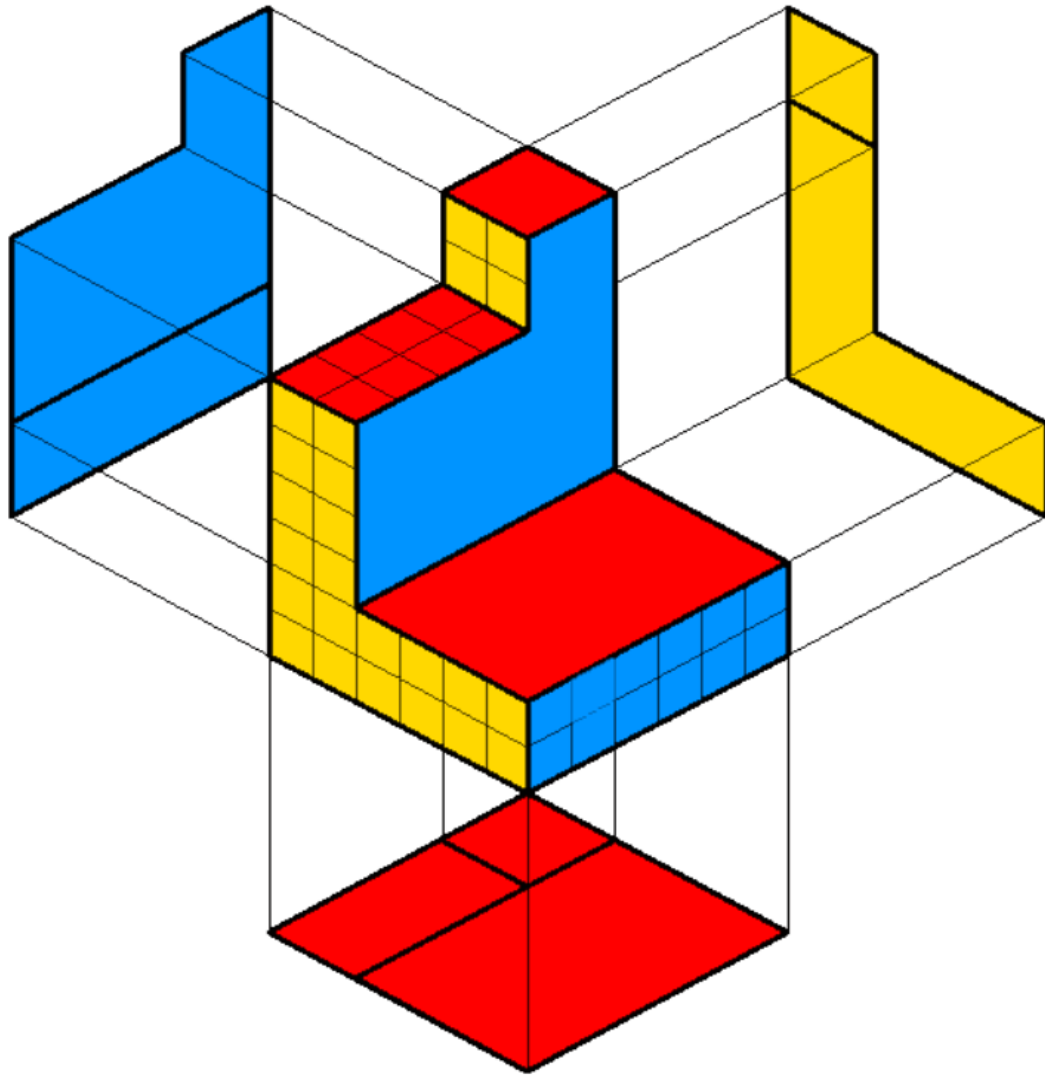


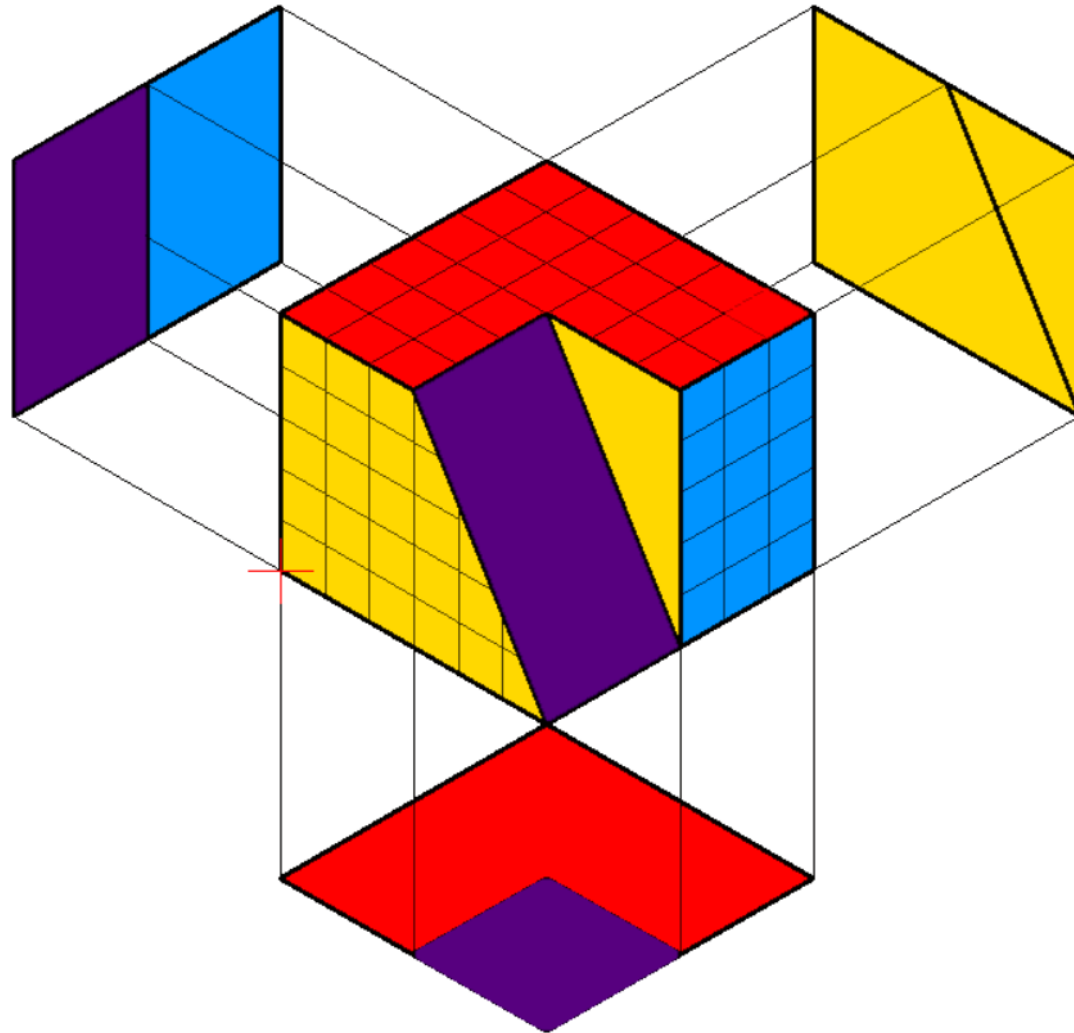
3. Bölge Sistemi

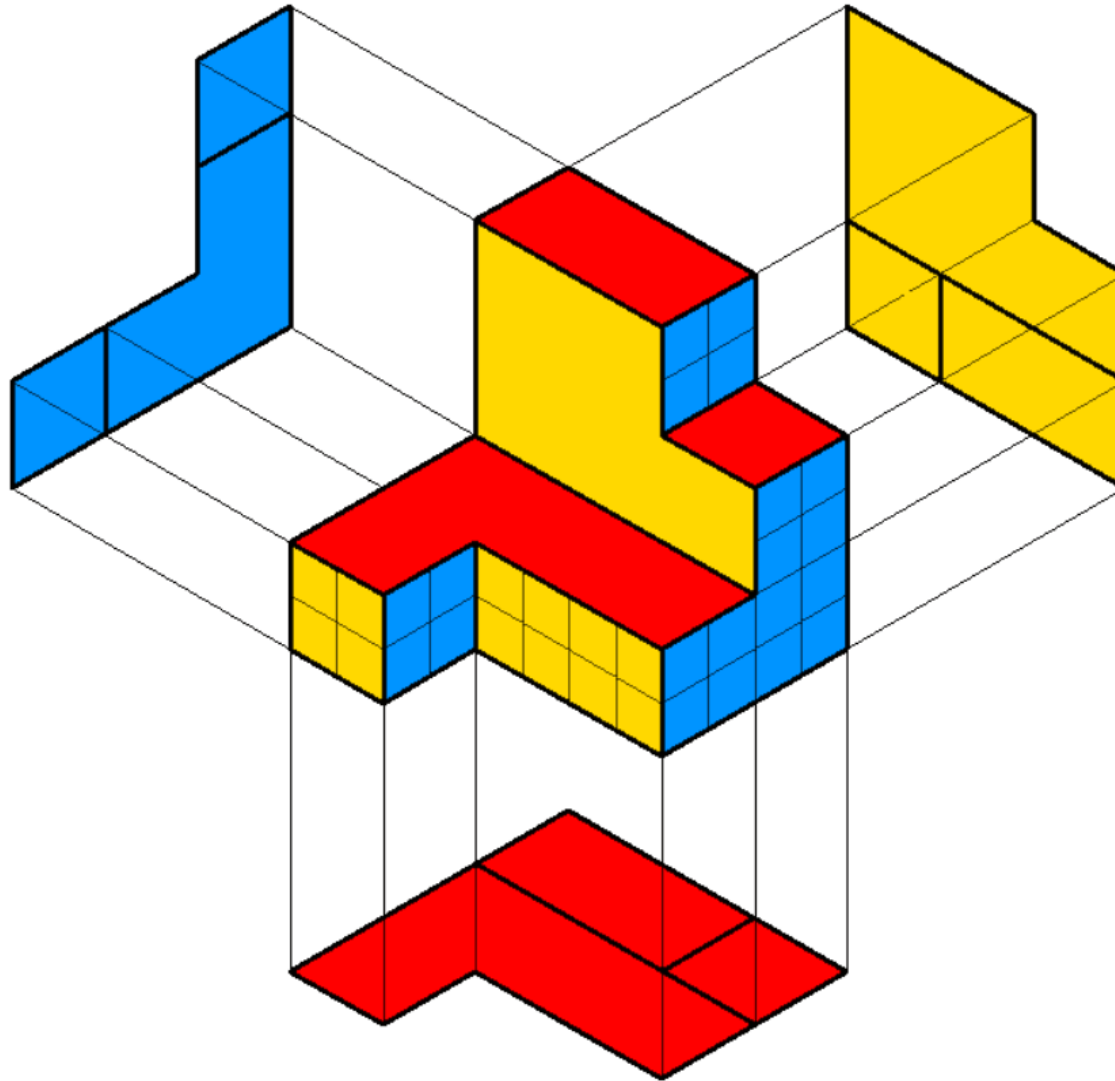


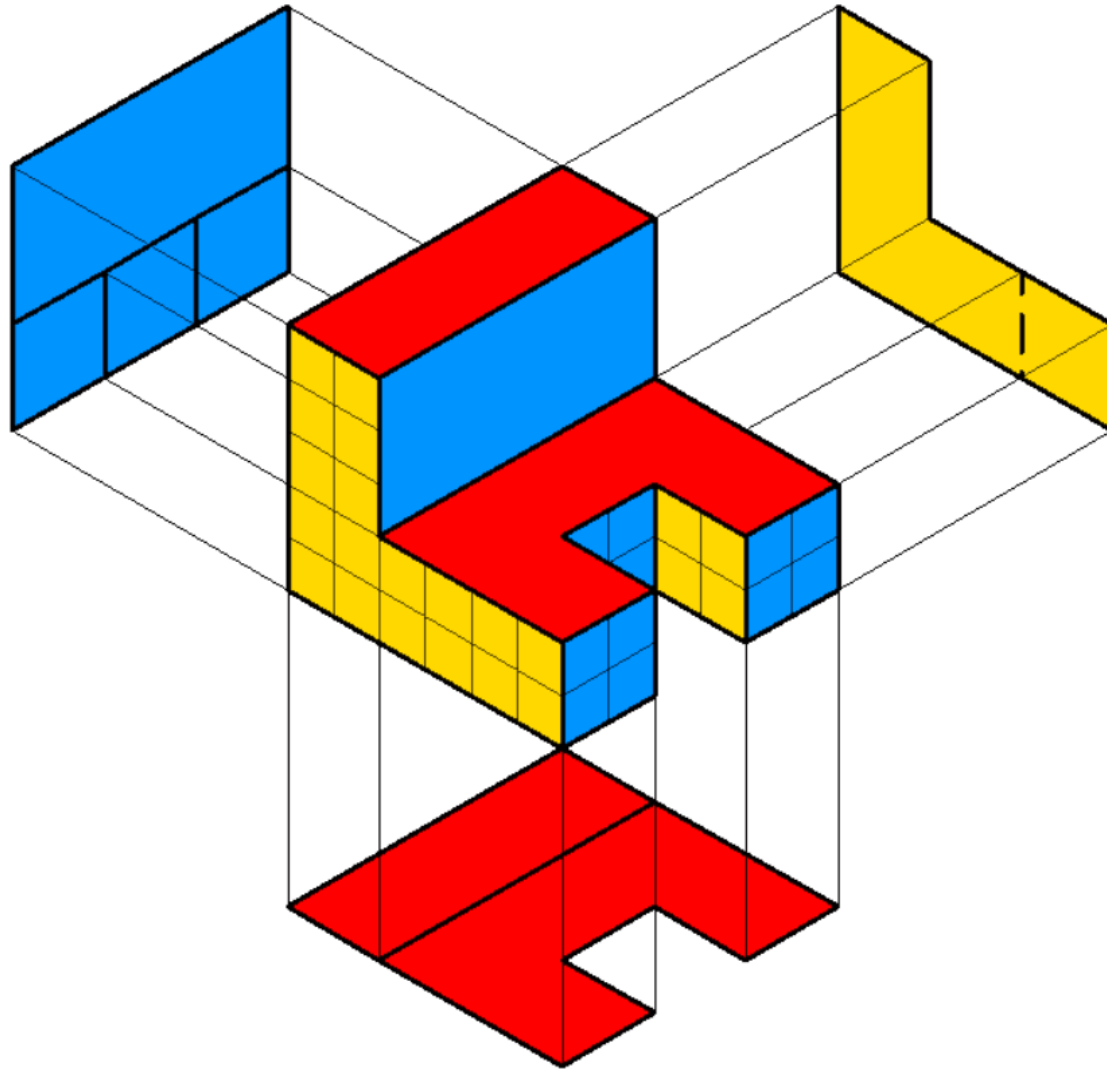


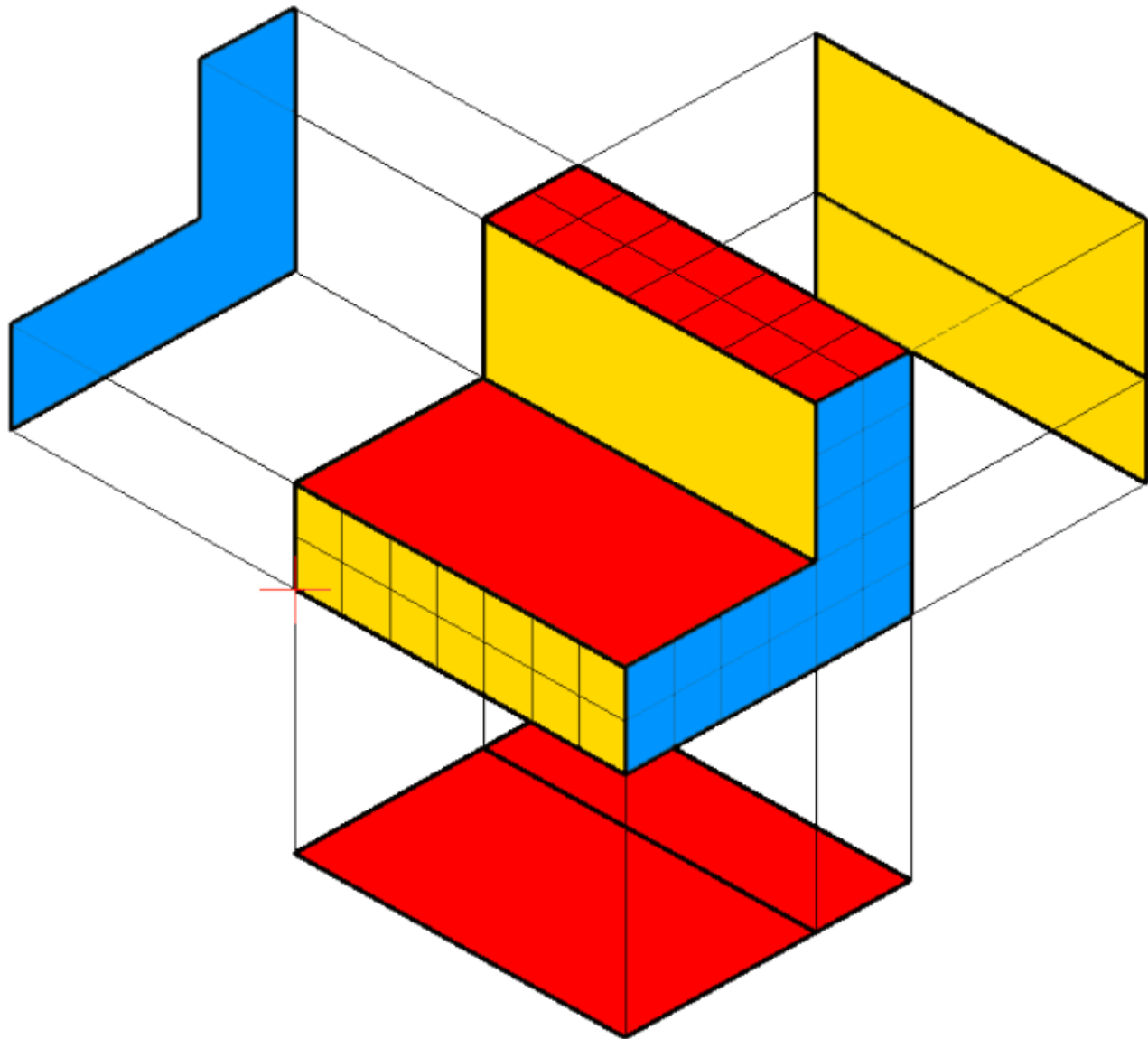
1.Bölge Sisteme Göre Standart Görünüşler

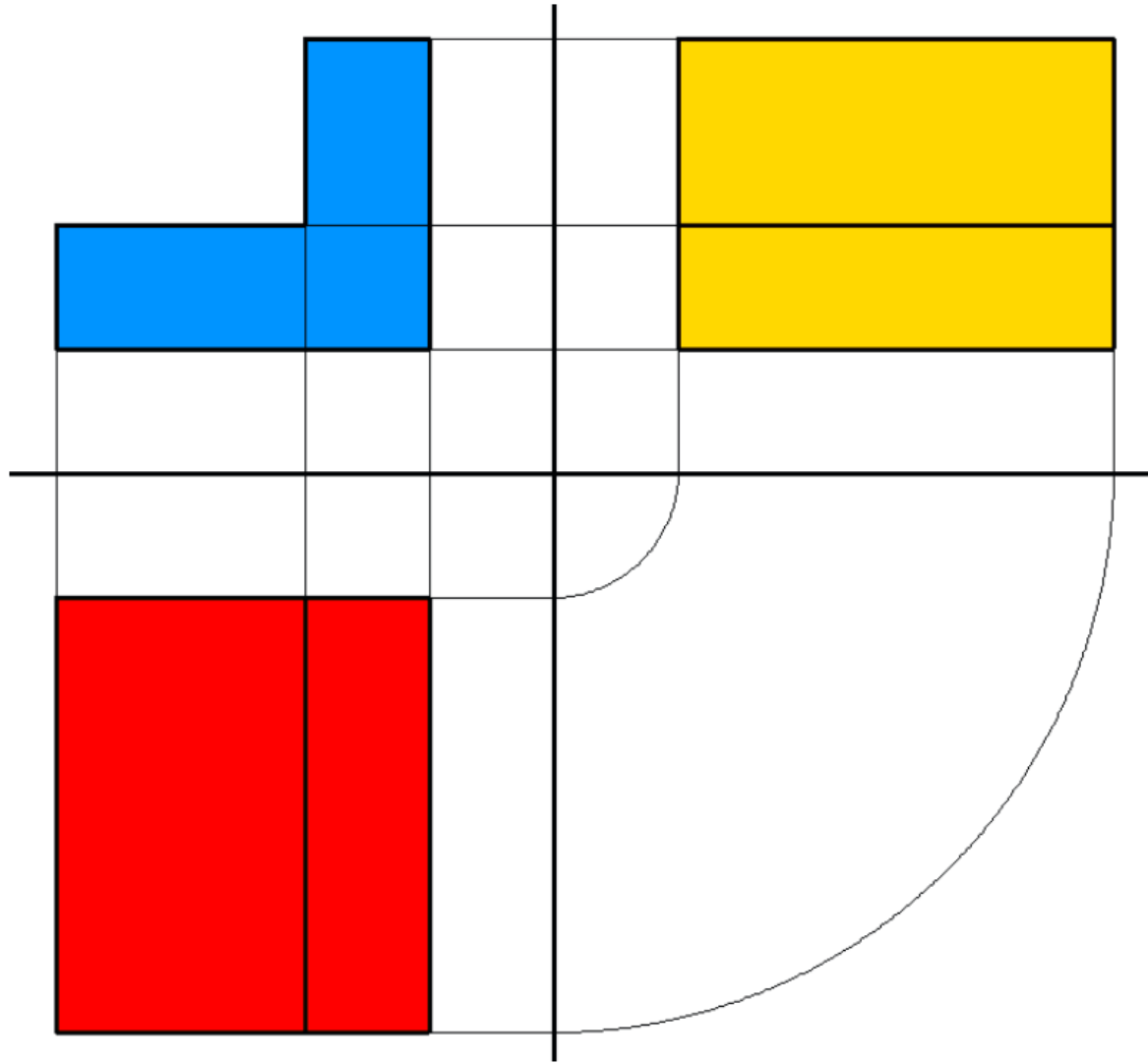


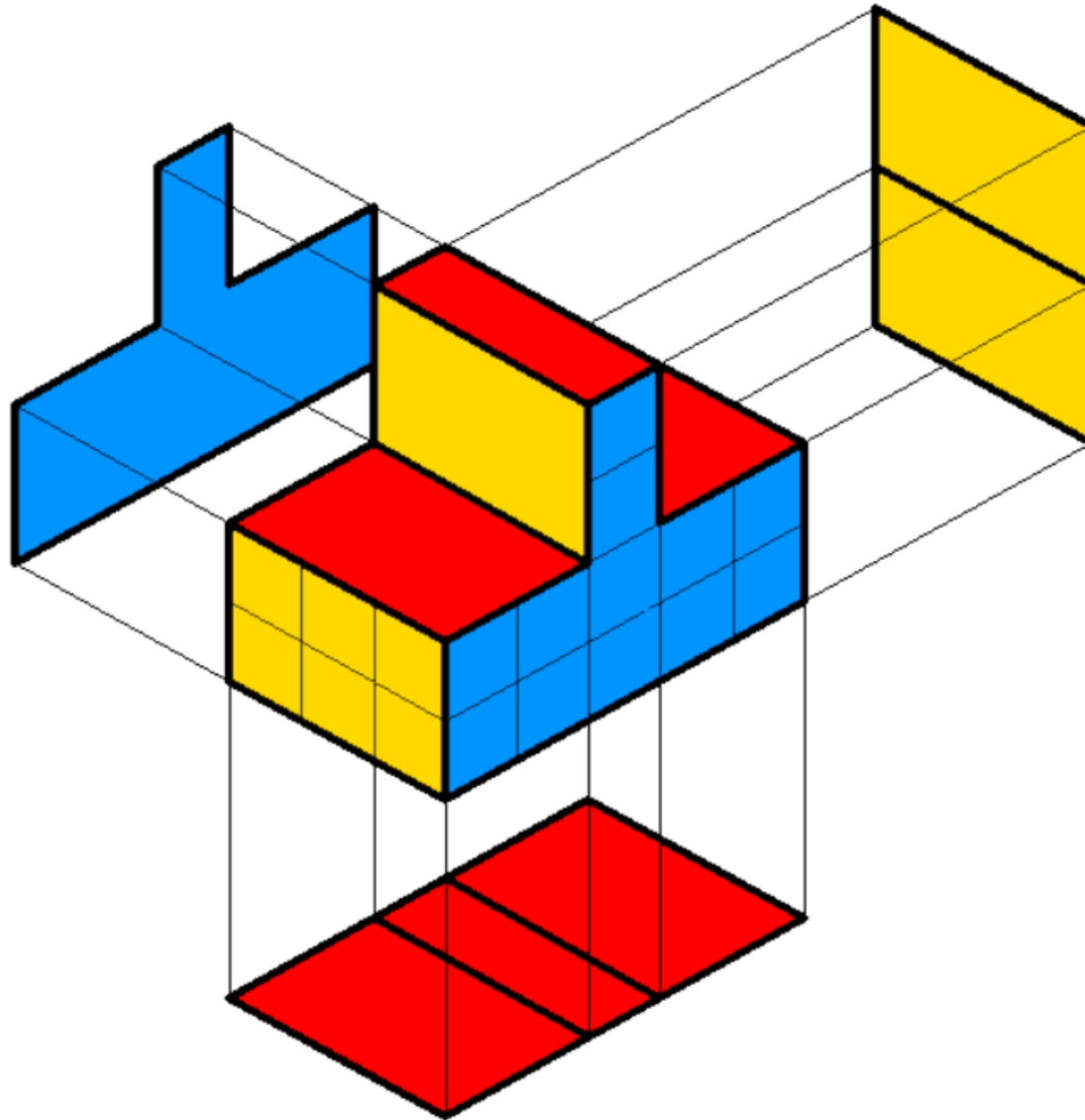


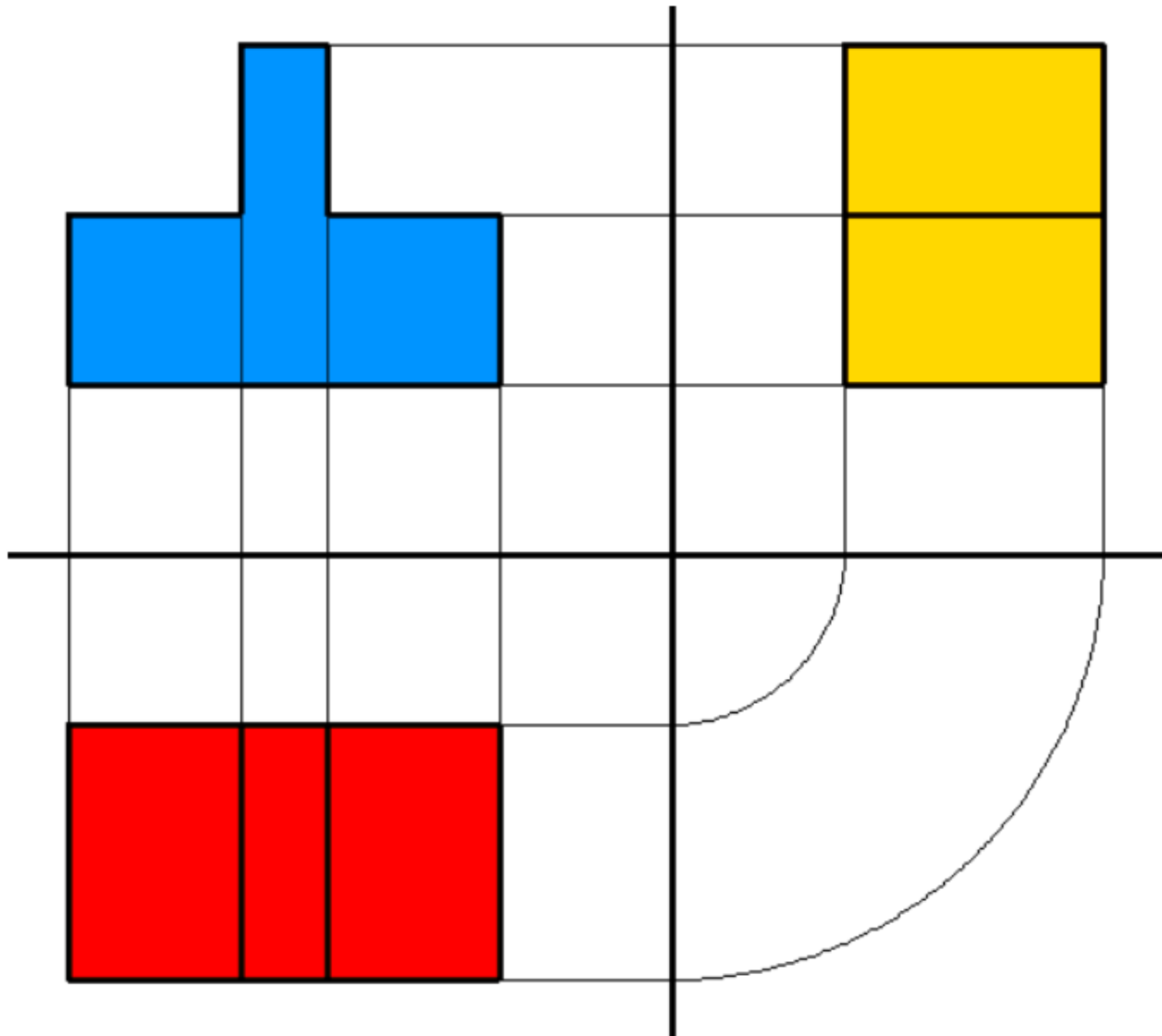


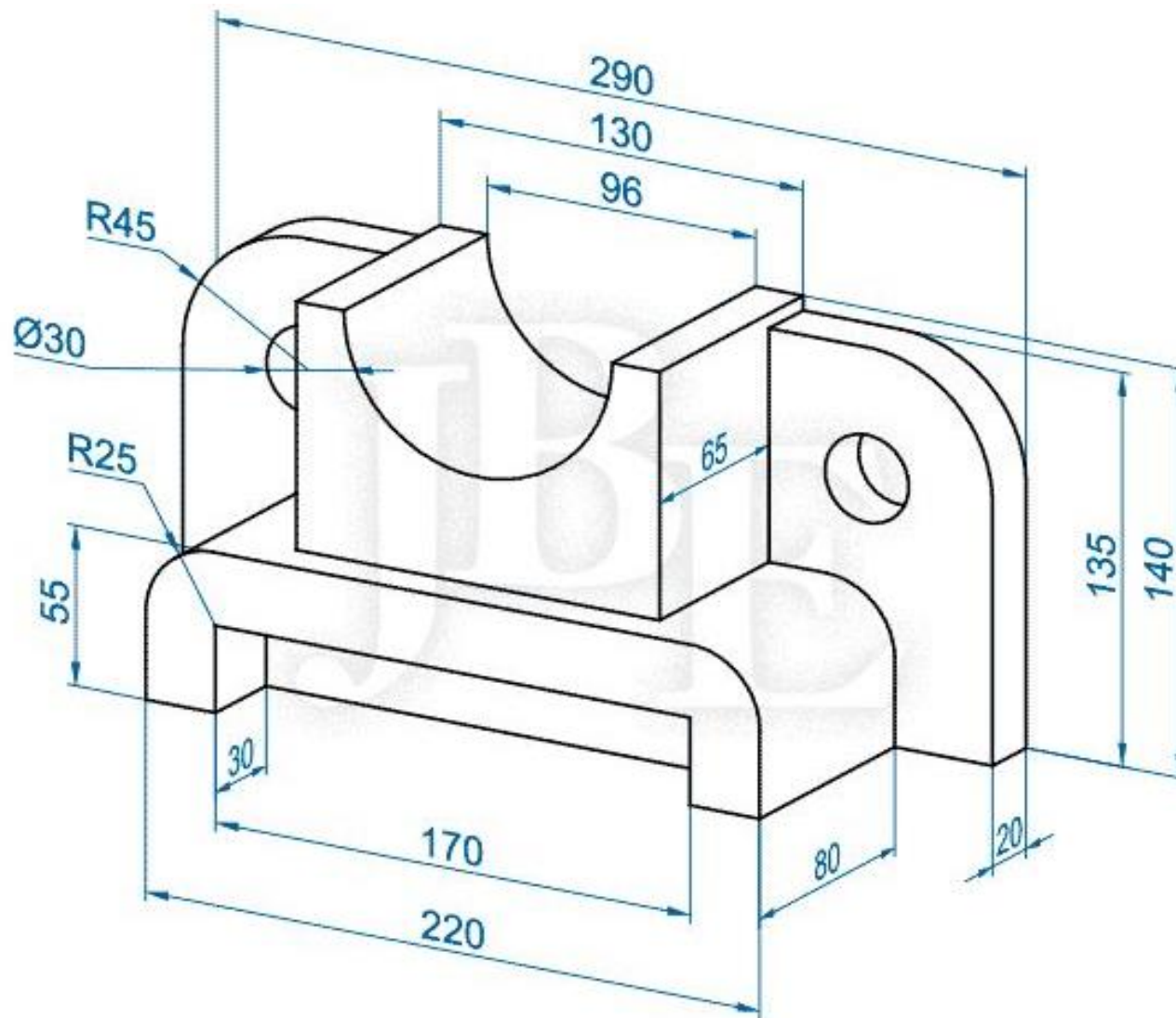


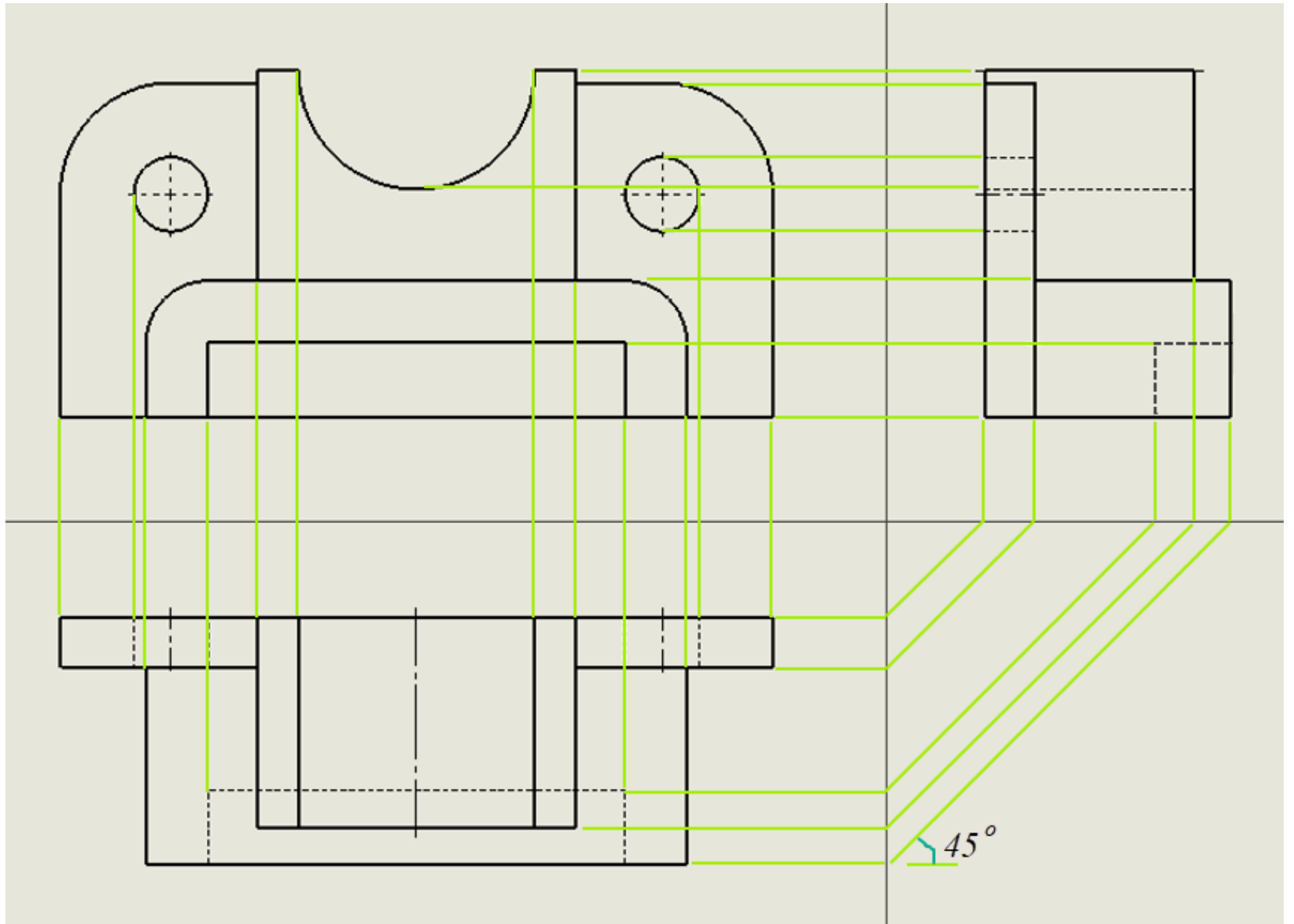


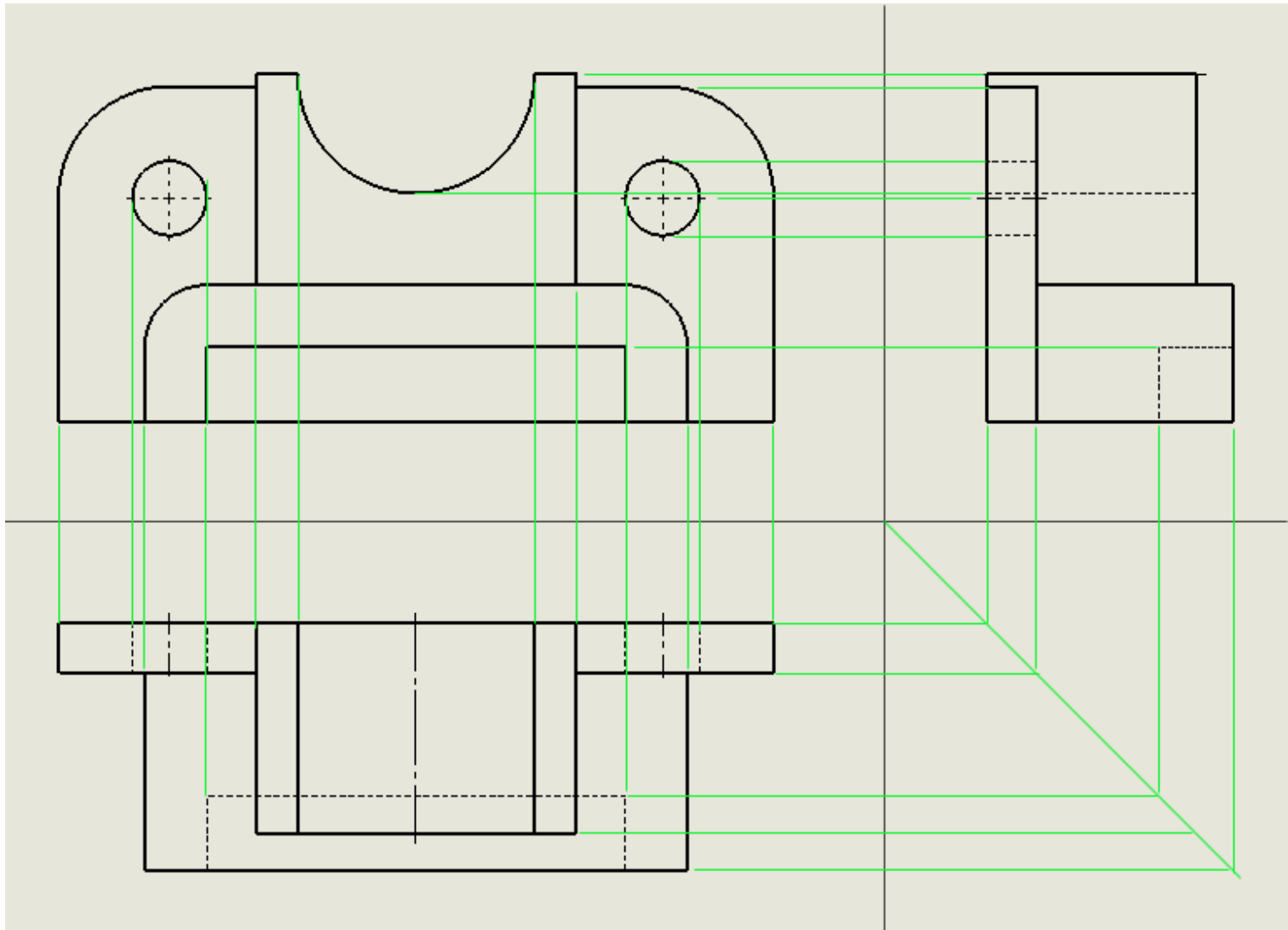


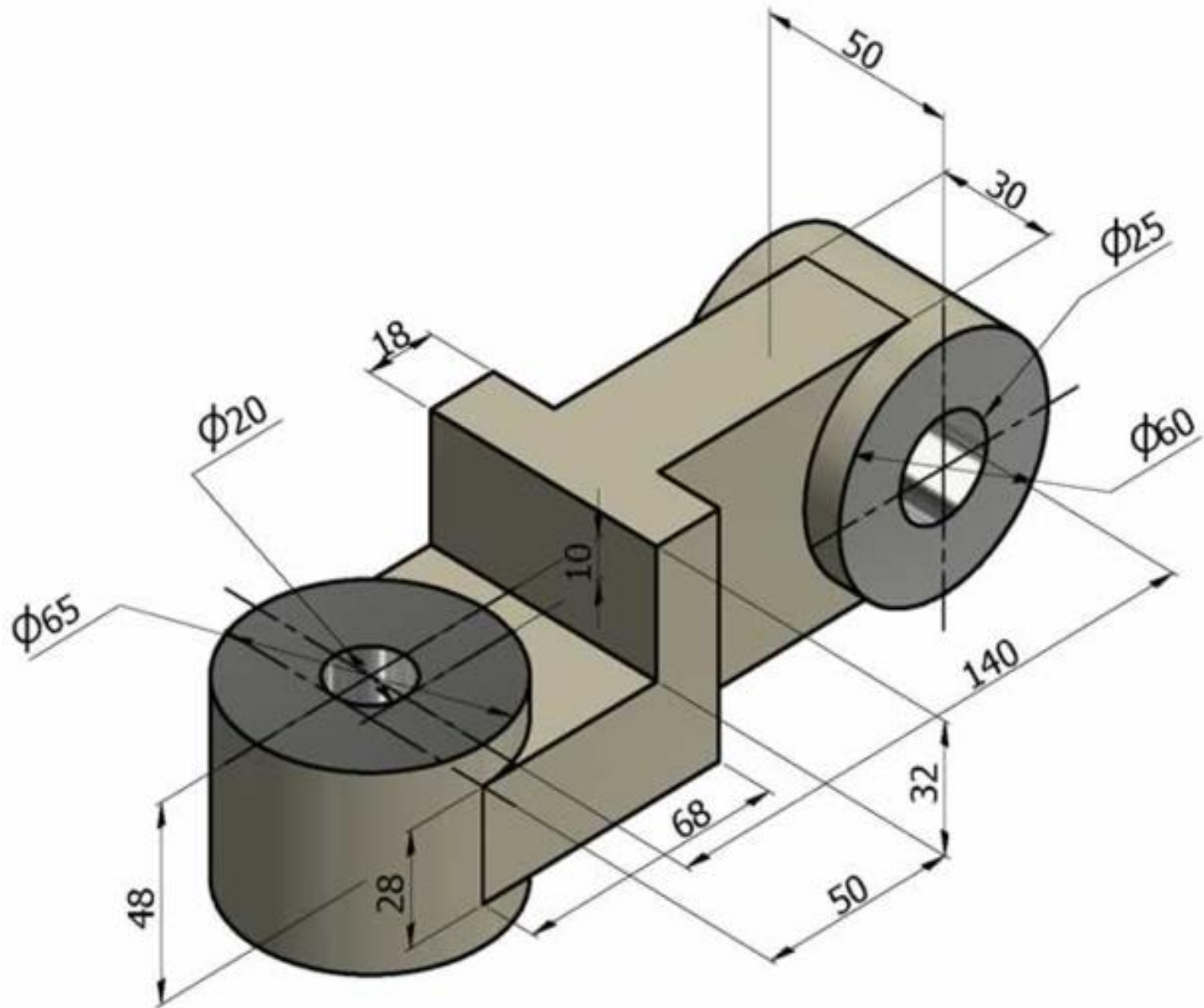


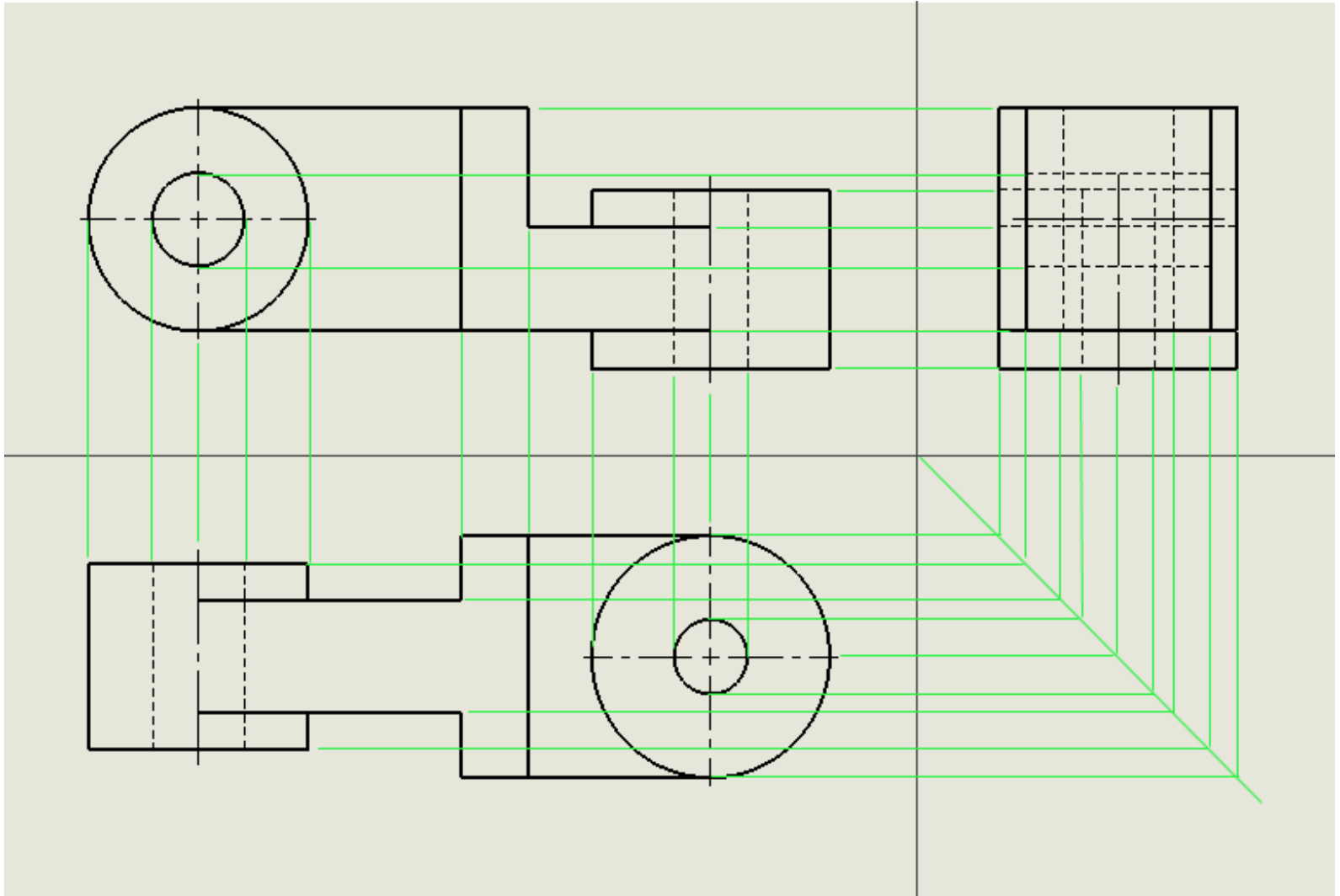


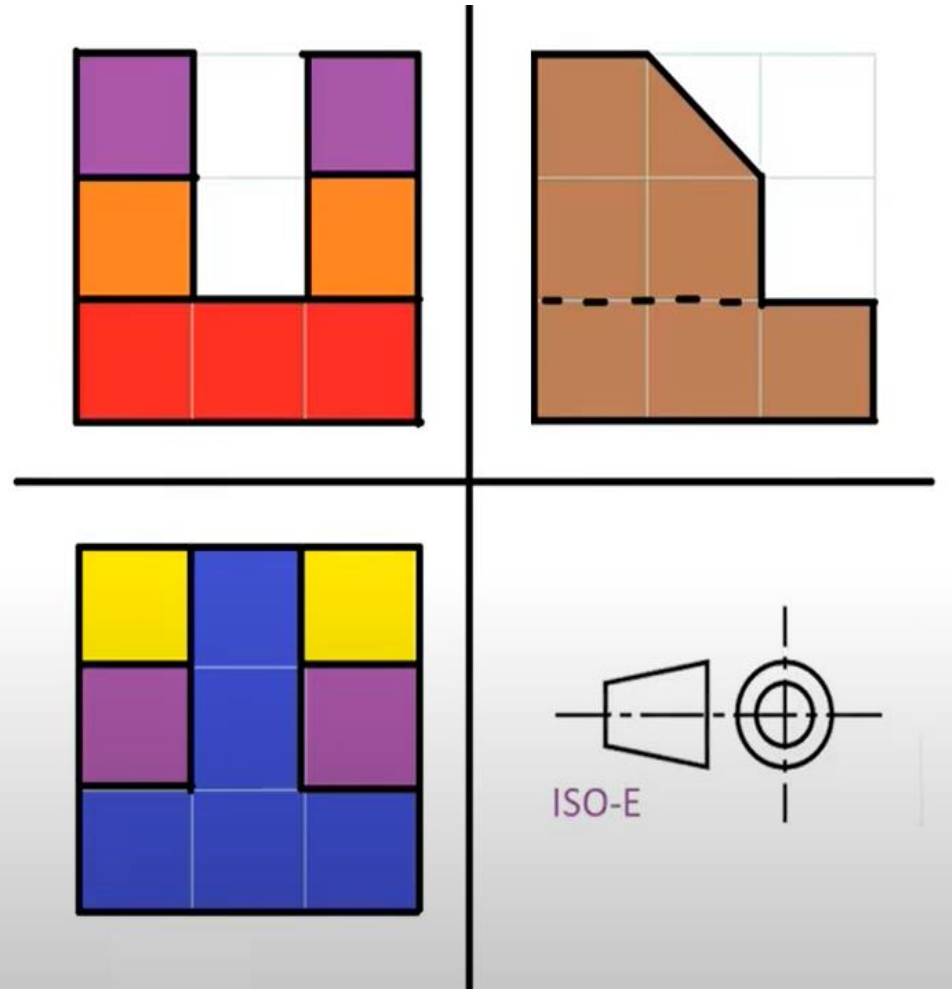
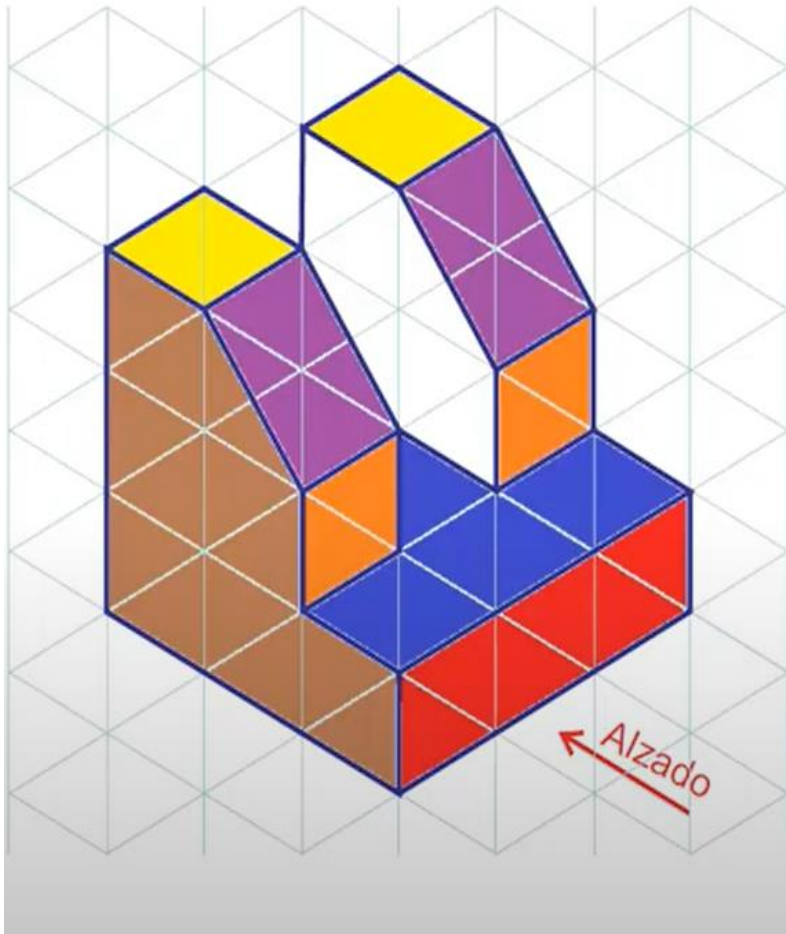


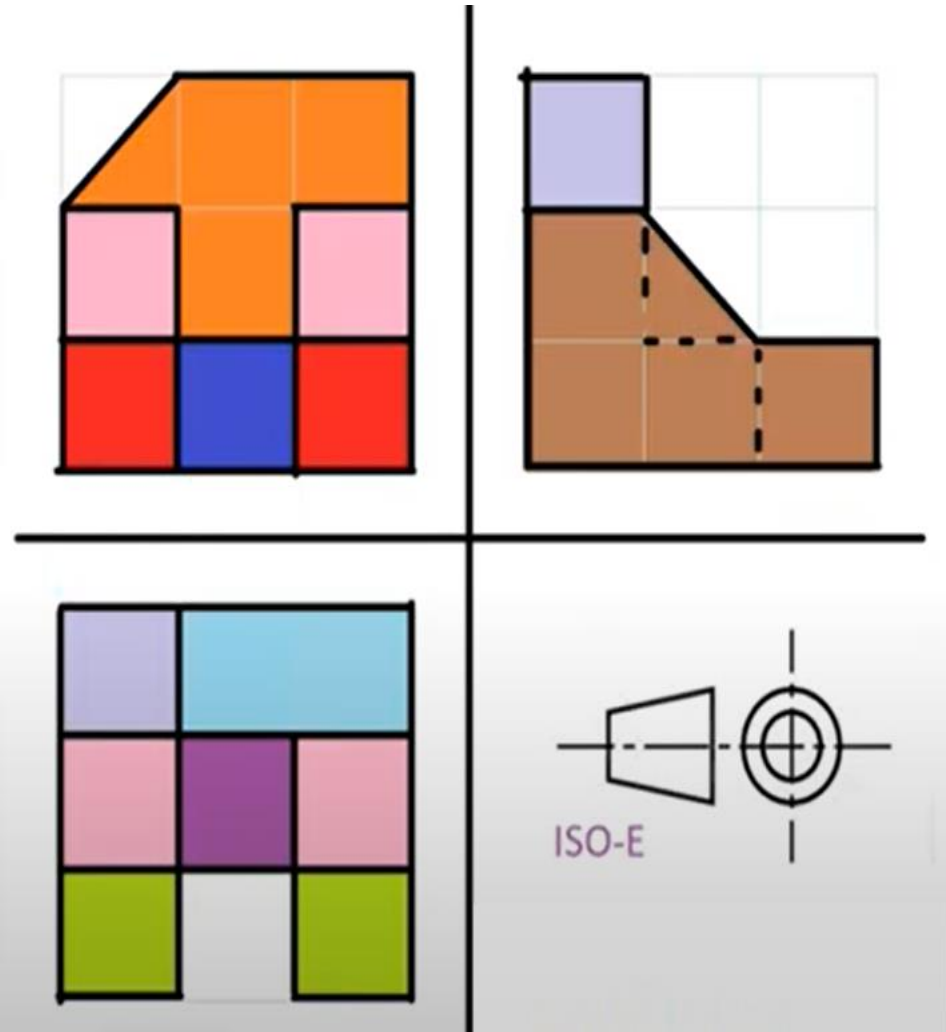
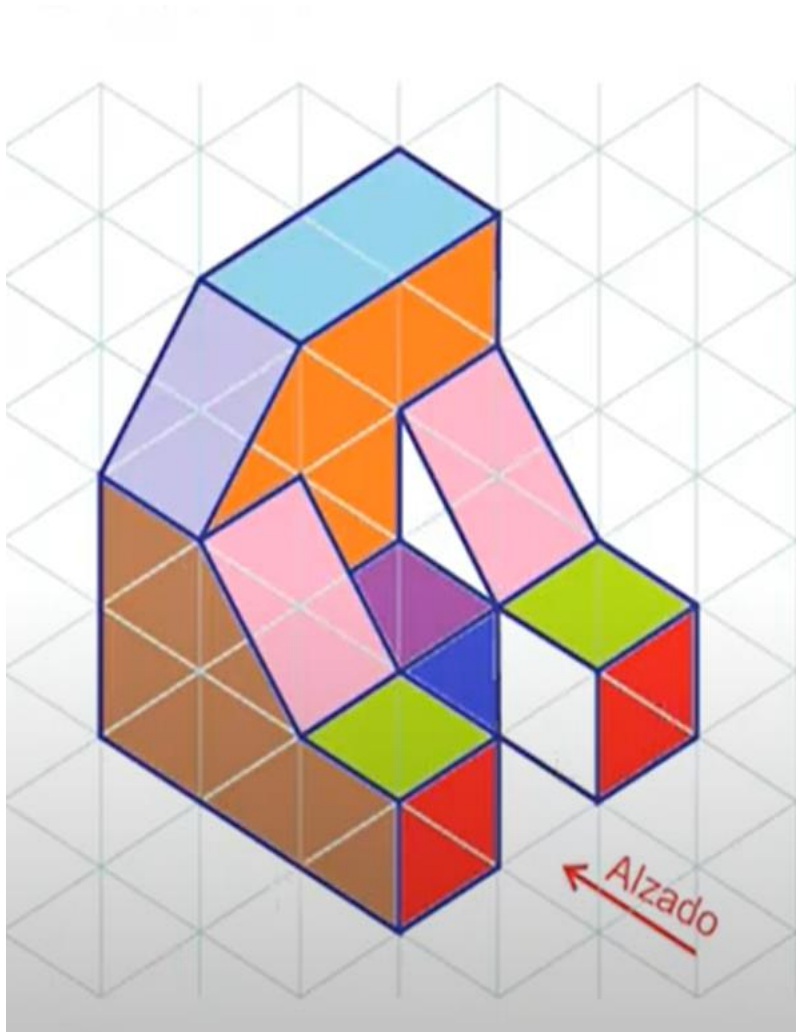


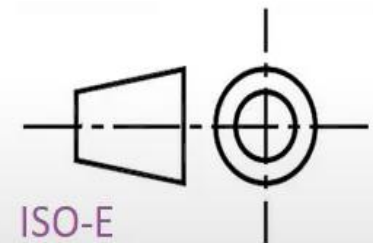
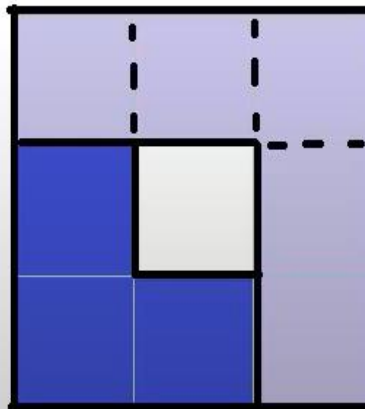
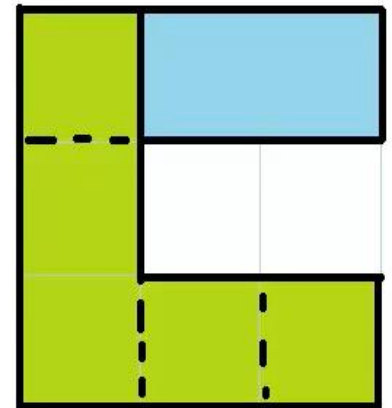
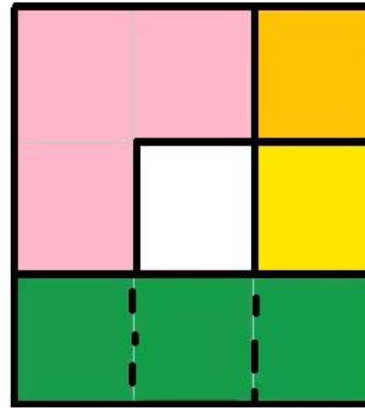
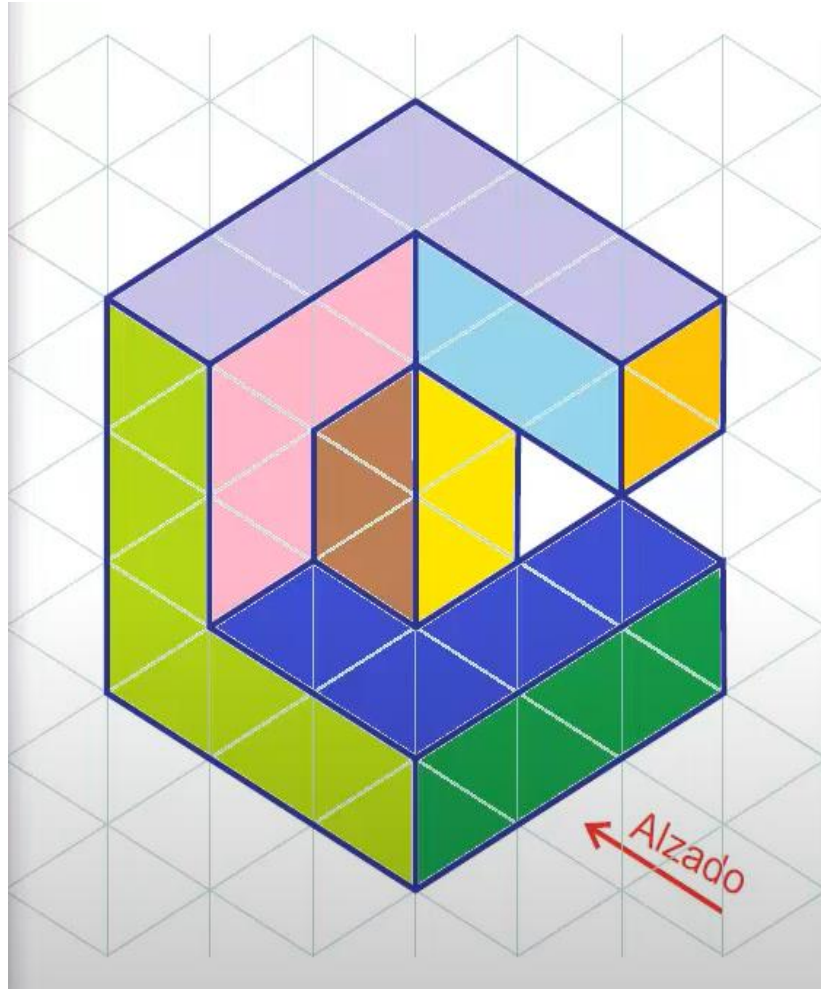


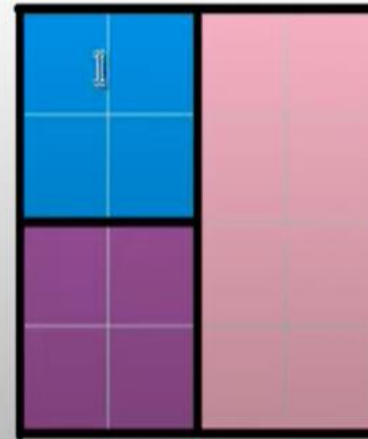
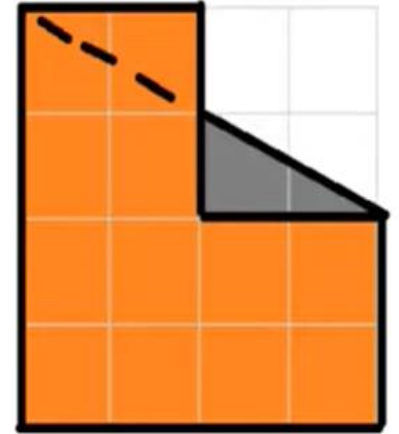
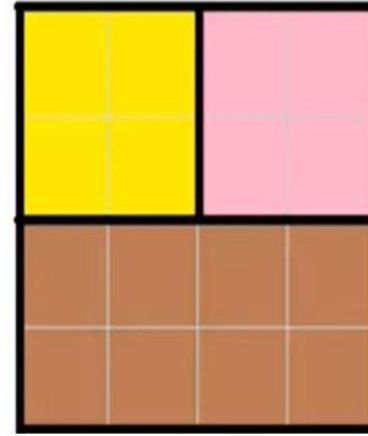
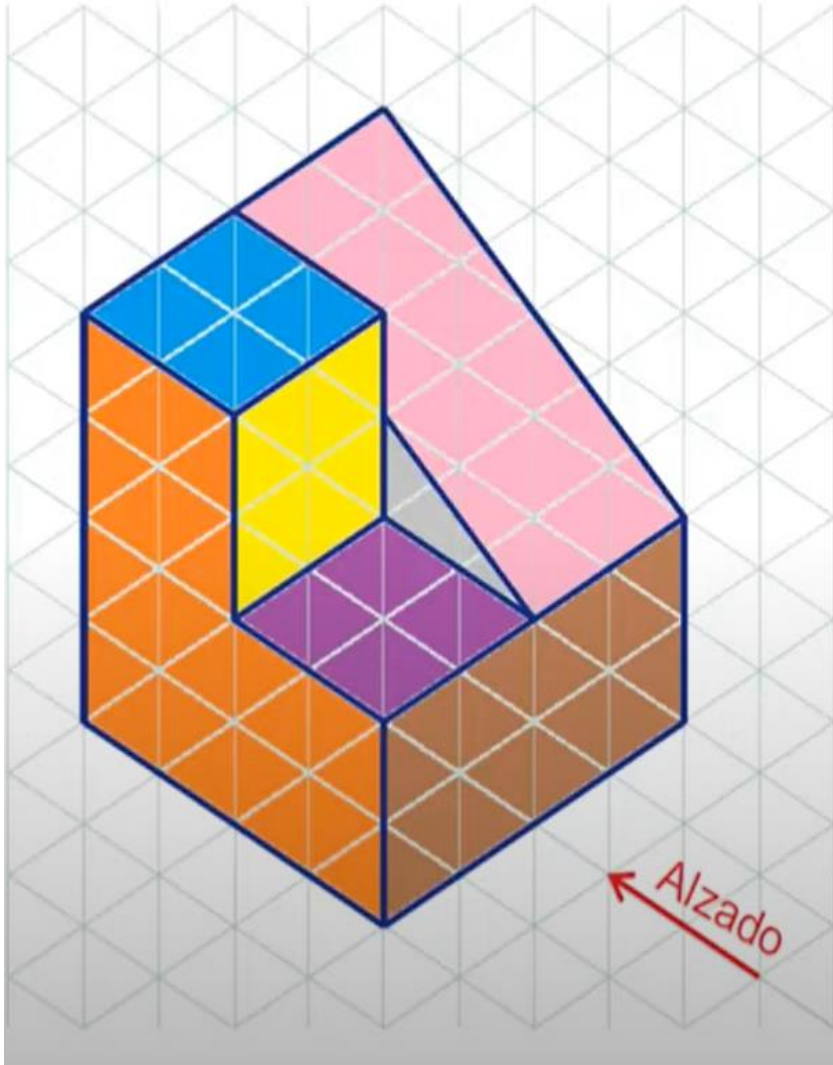


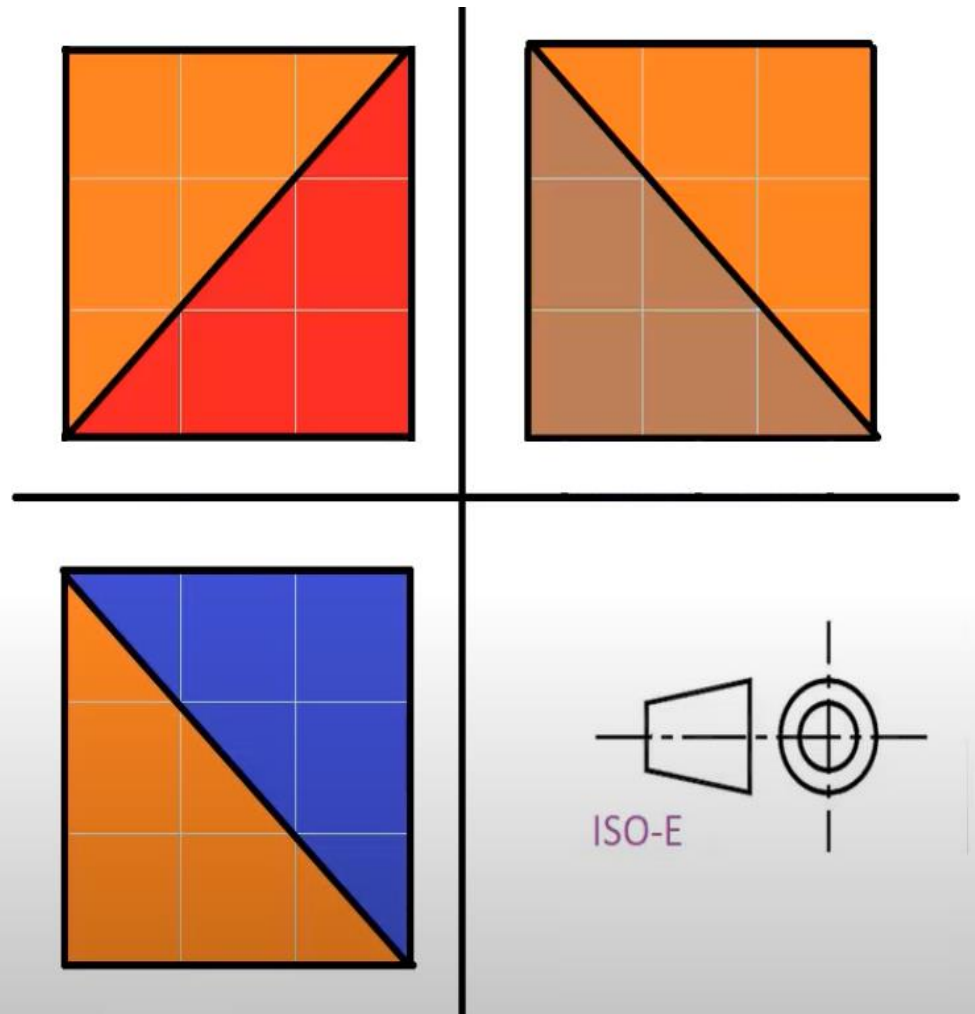
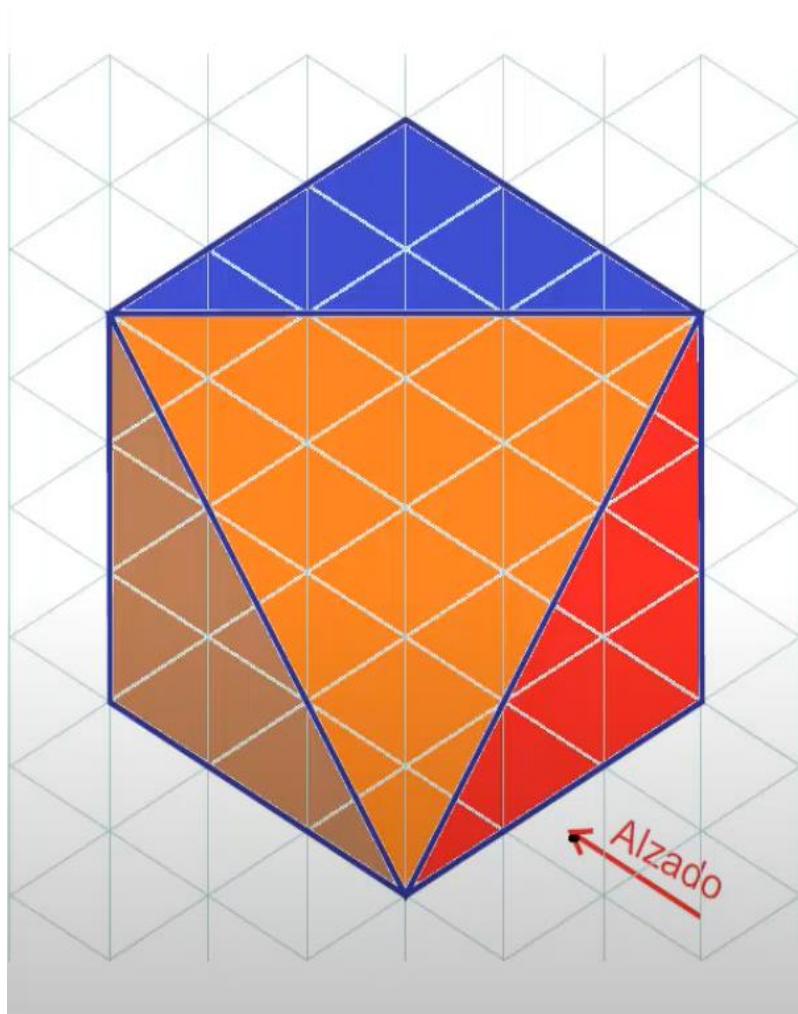




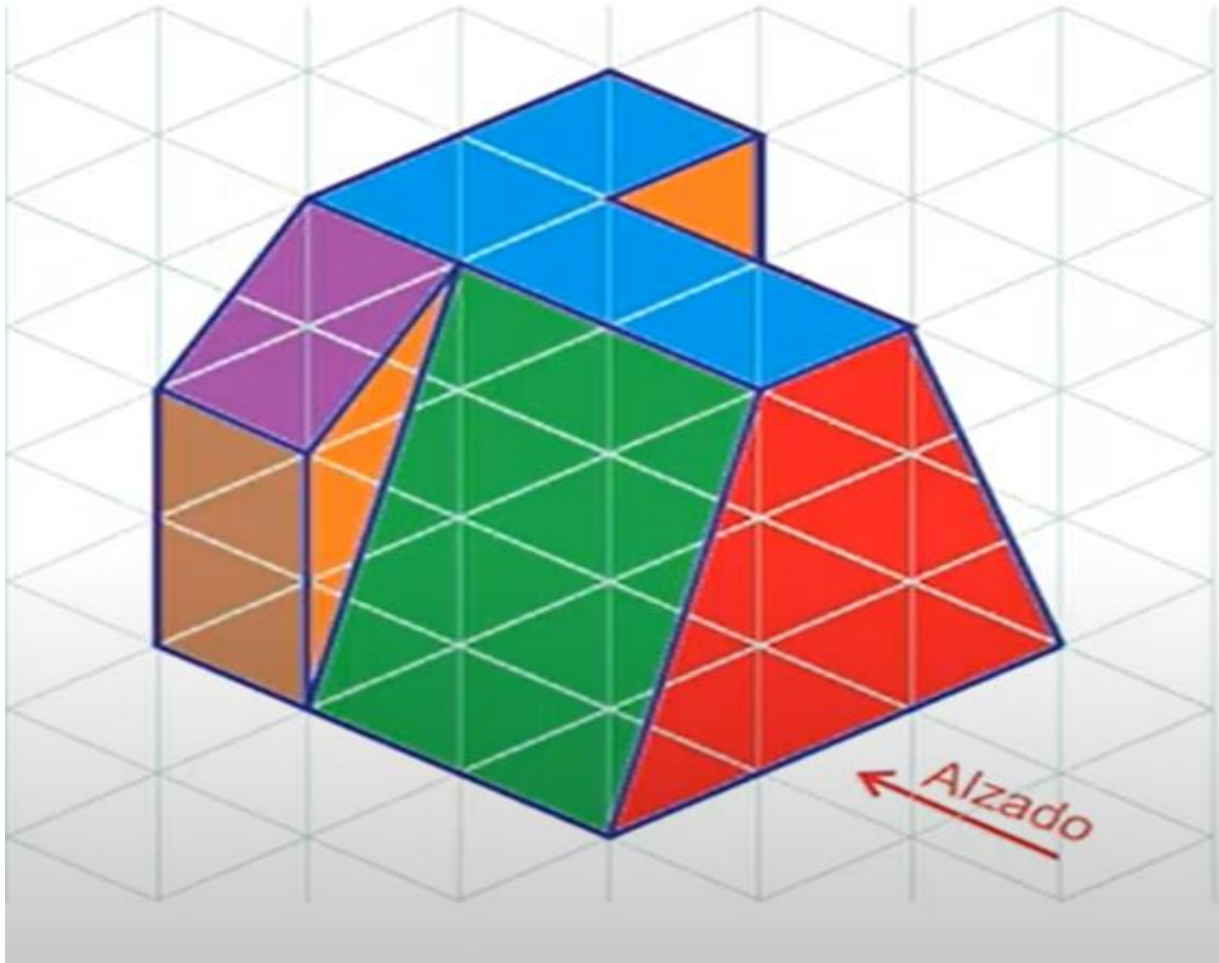




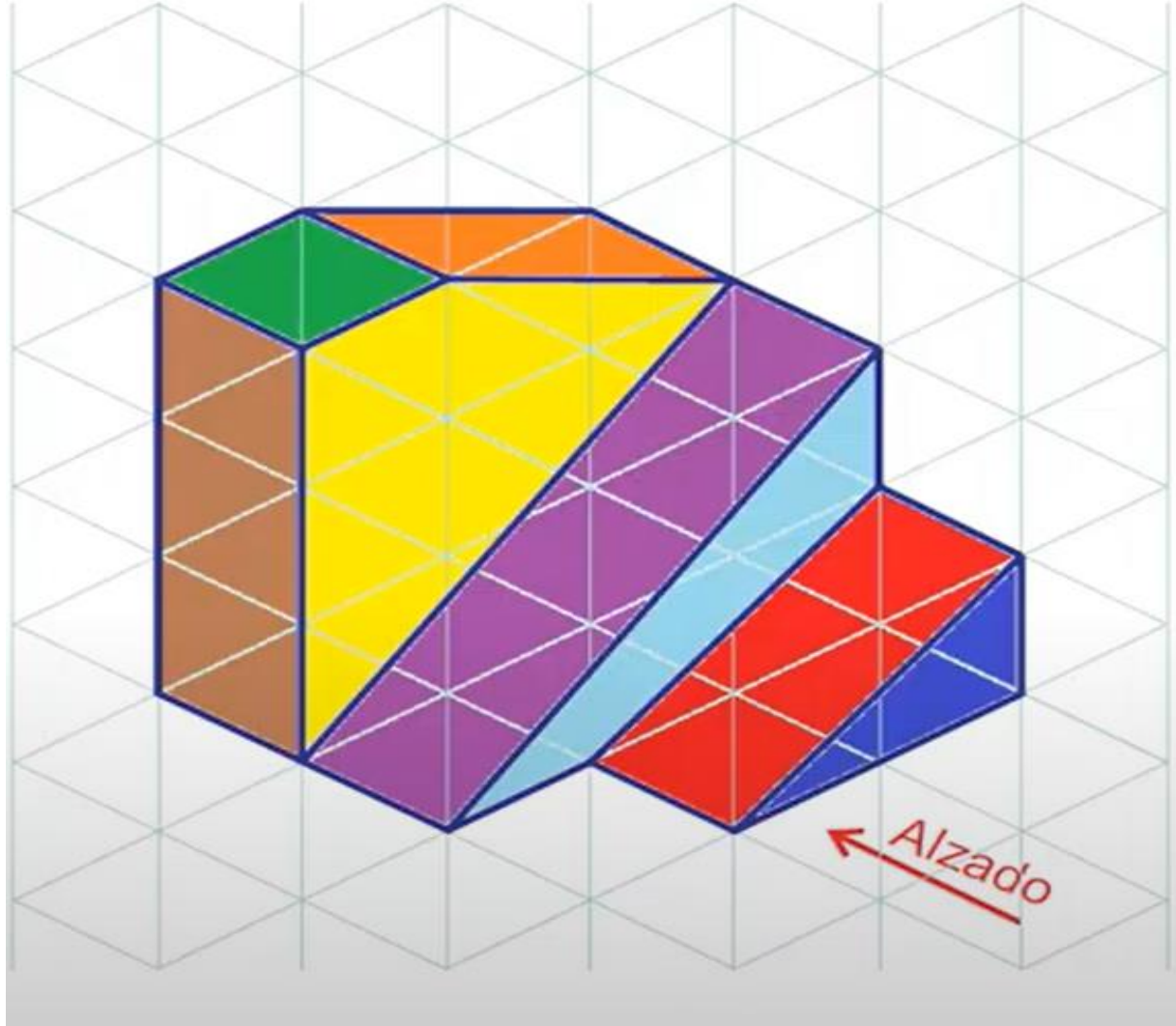




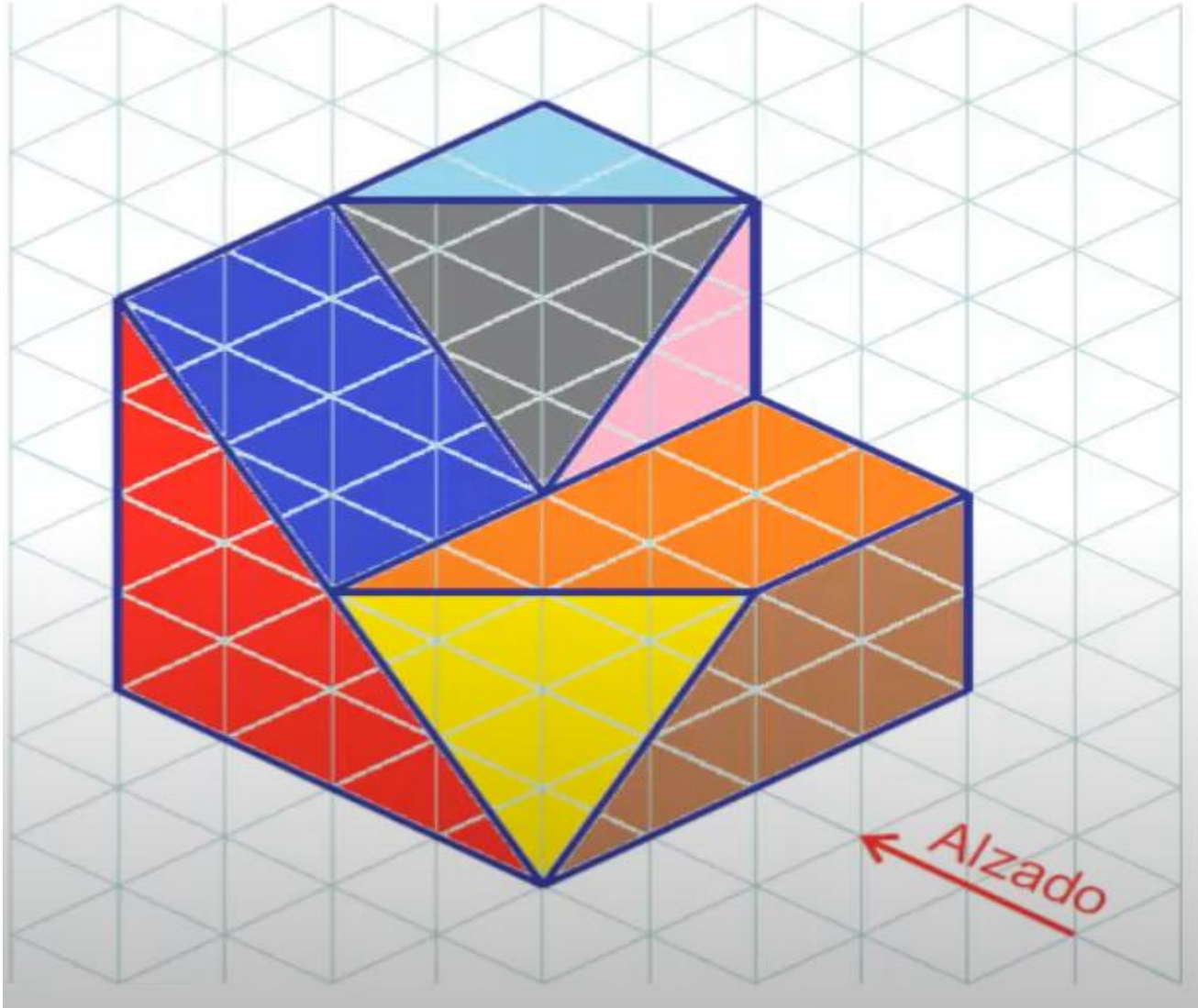
Ödev 1.



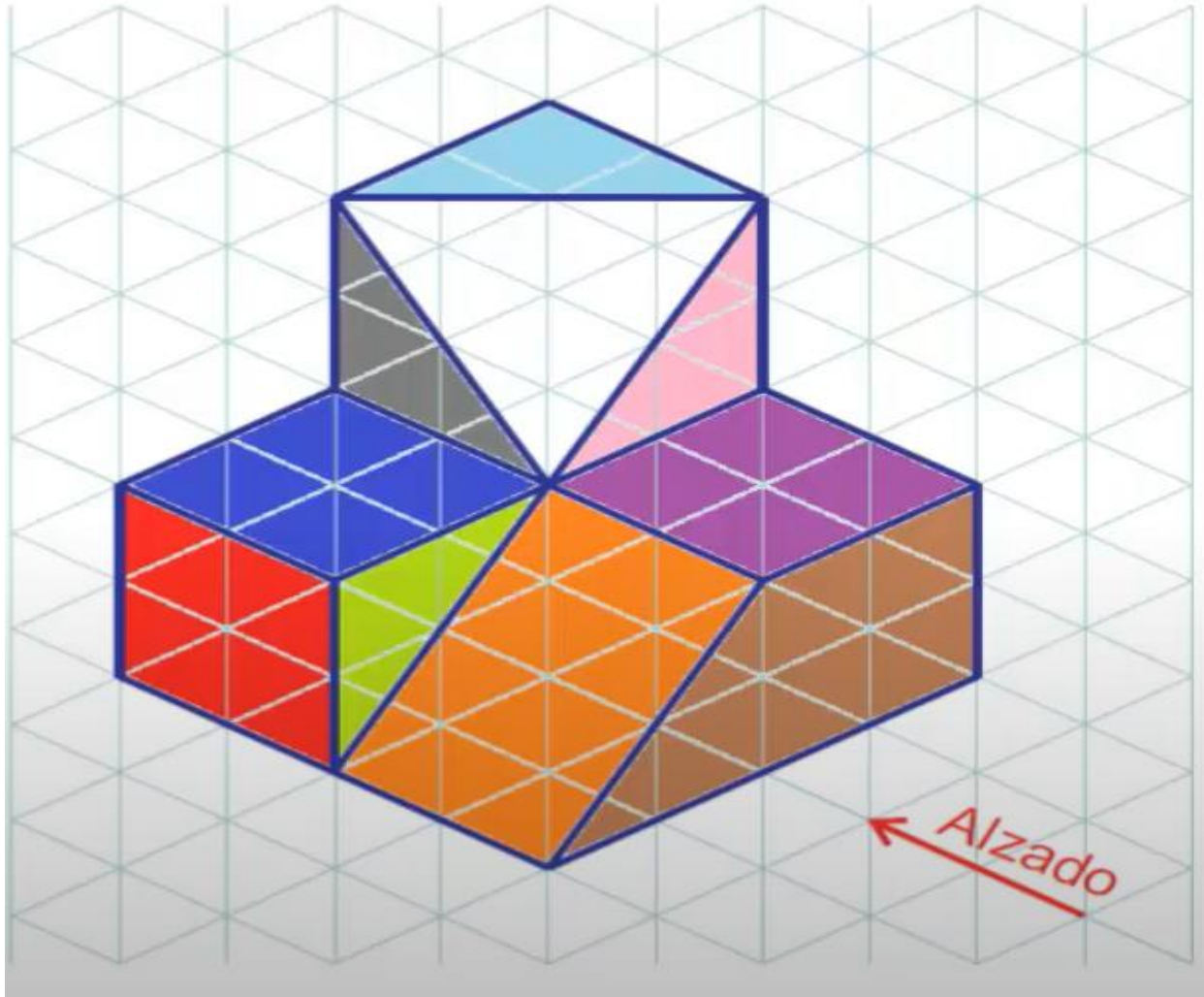
Ödev 2.



Ödev 3.



Ödev 4.

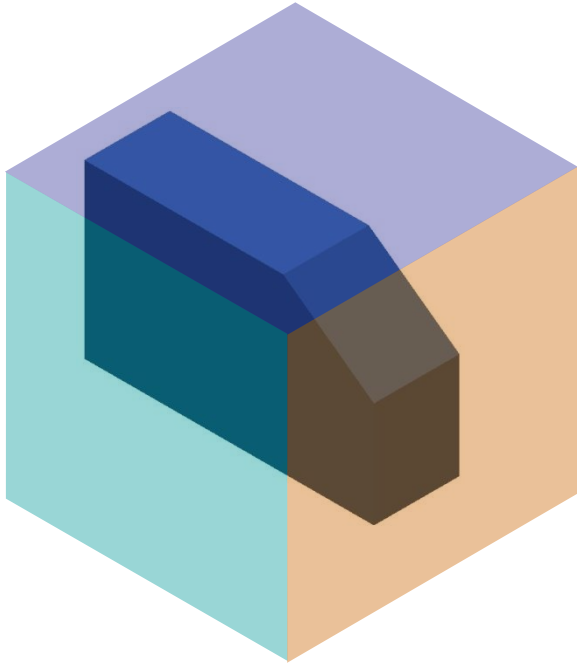




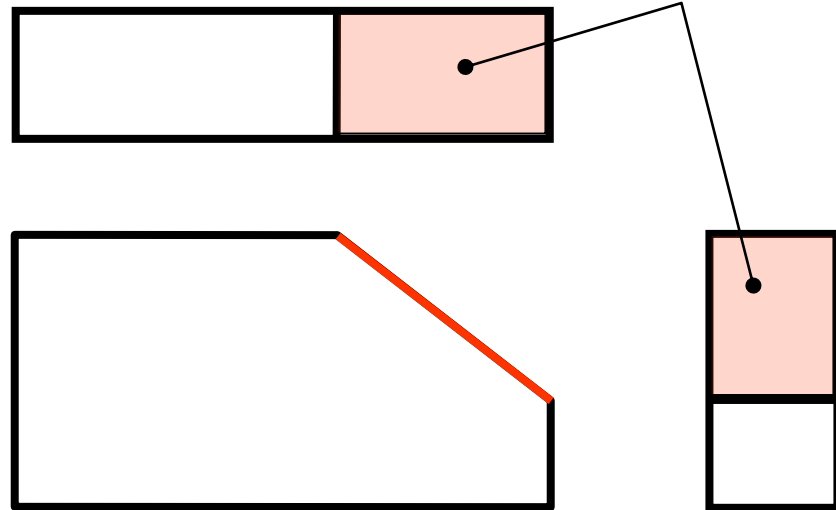
Yardımcı Görünüşler

Gerekliklik

- **Yardımcı Görünüşler** Asal izdüşüm düzlemlerinden herhangi birine paralel olmayan yüzeylerin gerçek boyut ve gerçek şeklini görmek için gereklidir.
-



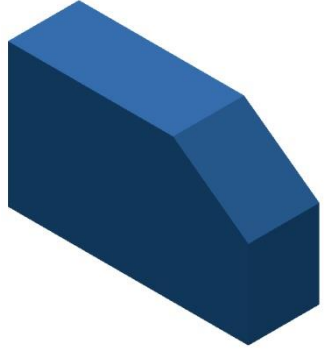
Bu görüşlerden gerçek boyut alınamaz



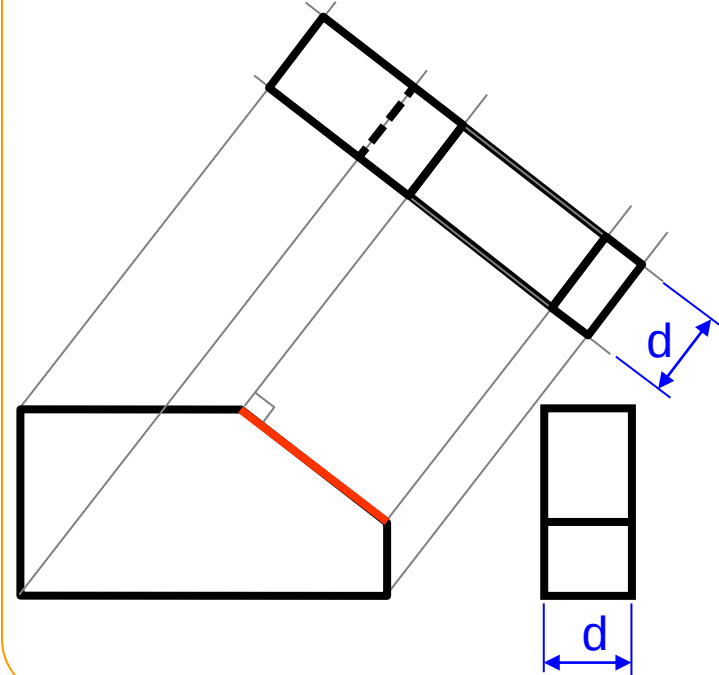
Yardımcı Görünüş

- Pratikte, yardımcı görünüşler sadece istenilen bilgileri gösterecek **kısmi görünüş** olarak çıkartılır.

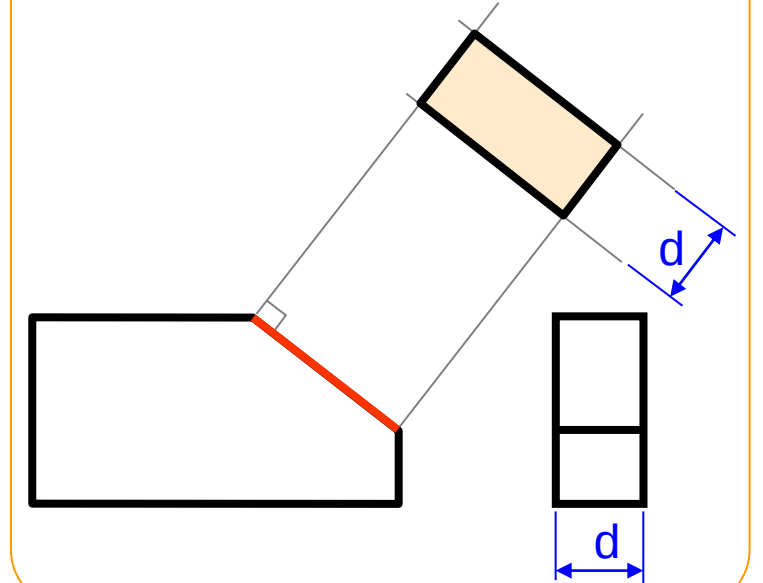
Örnek



Komple Y.Görünüş



Kısmi Y. Görünüş



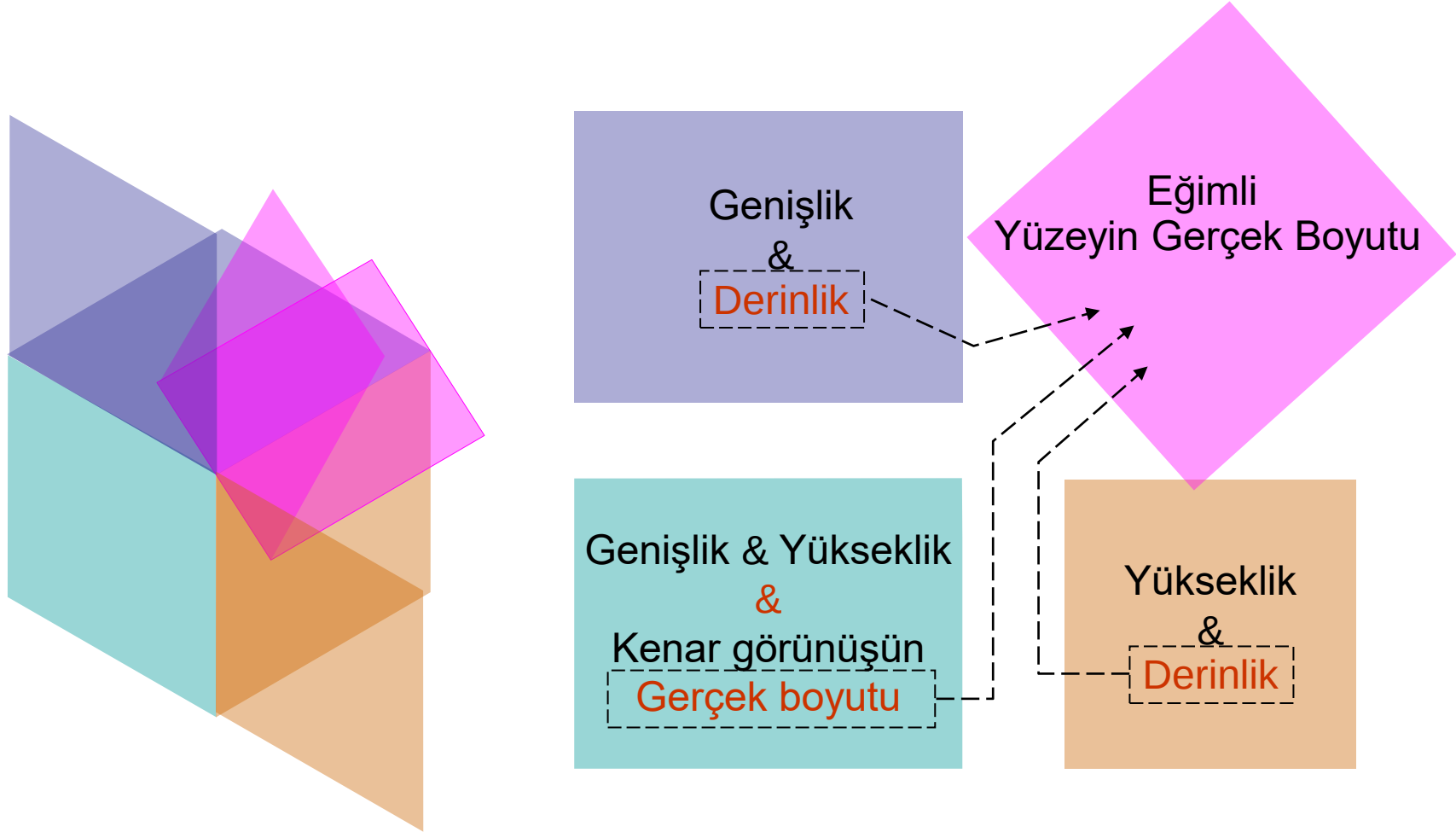
Yardımcı Görünüş Çeşitleri

■ Ana yardımcı görüşler 3 temel görüşe bağlı olarak sınıflandırılır.

1. Ön Görünüşe Komşu
2. Üst Görünüşe Komşu
3. Yan Görünüşe Komşu

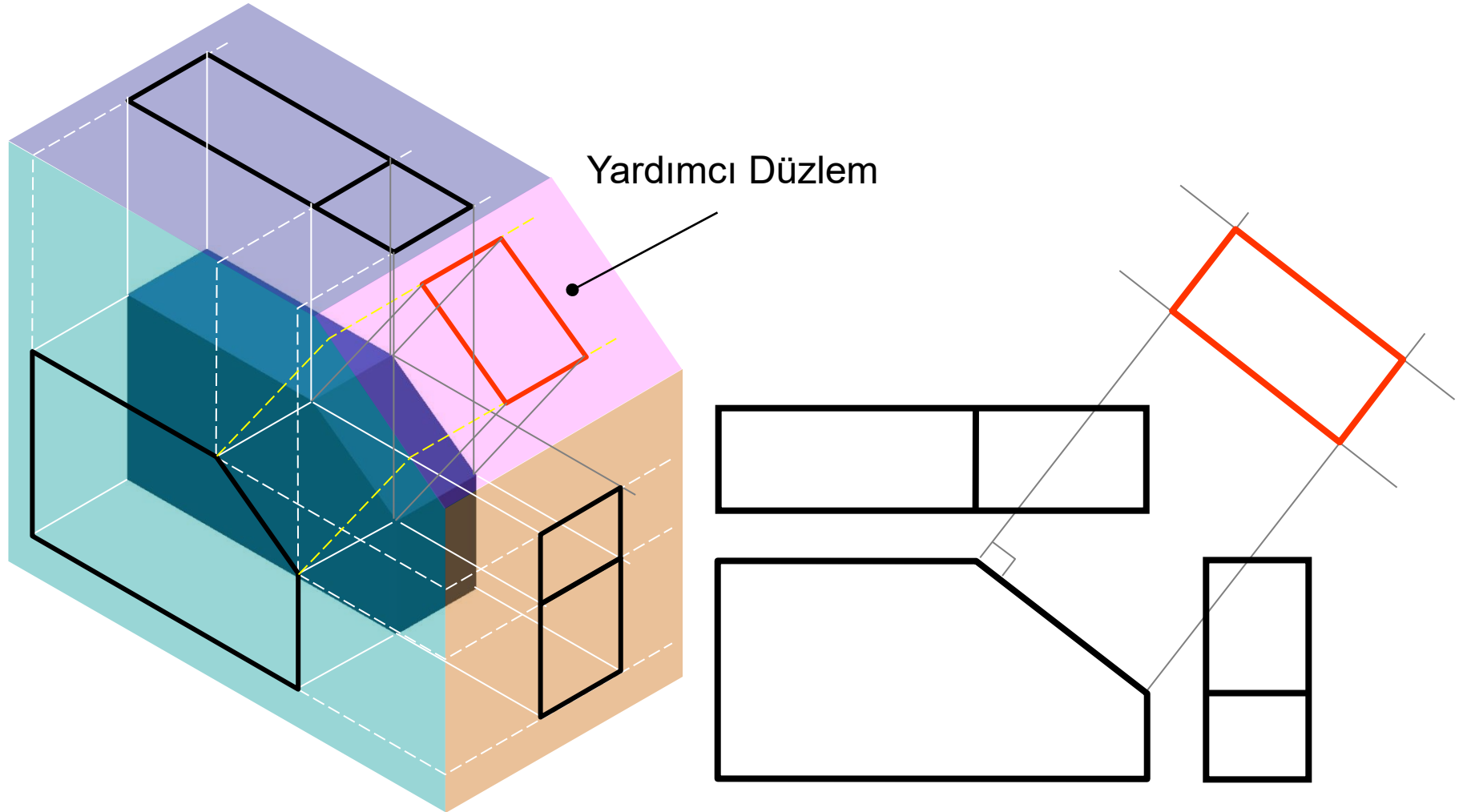
Ön Görünüşe Komşu Yardımcı Görünüşler

Cam Kutu modeli ve Döndürme Görünüş Düzenleme



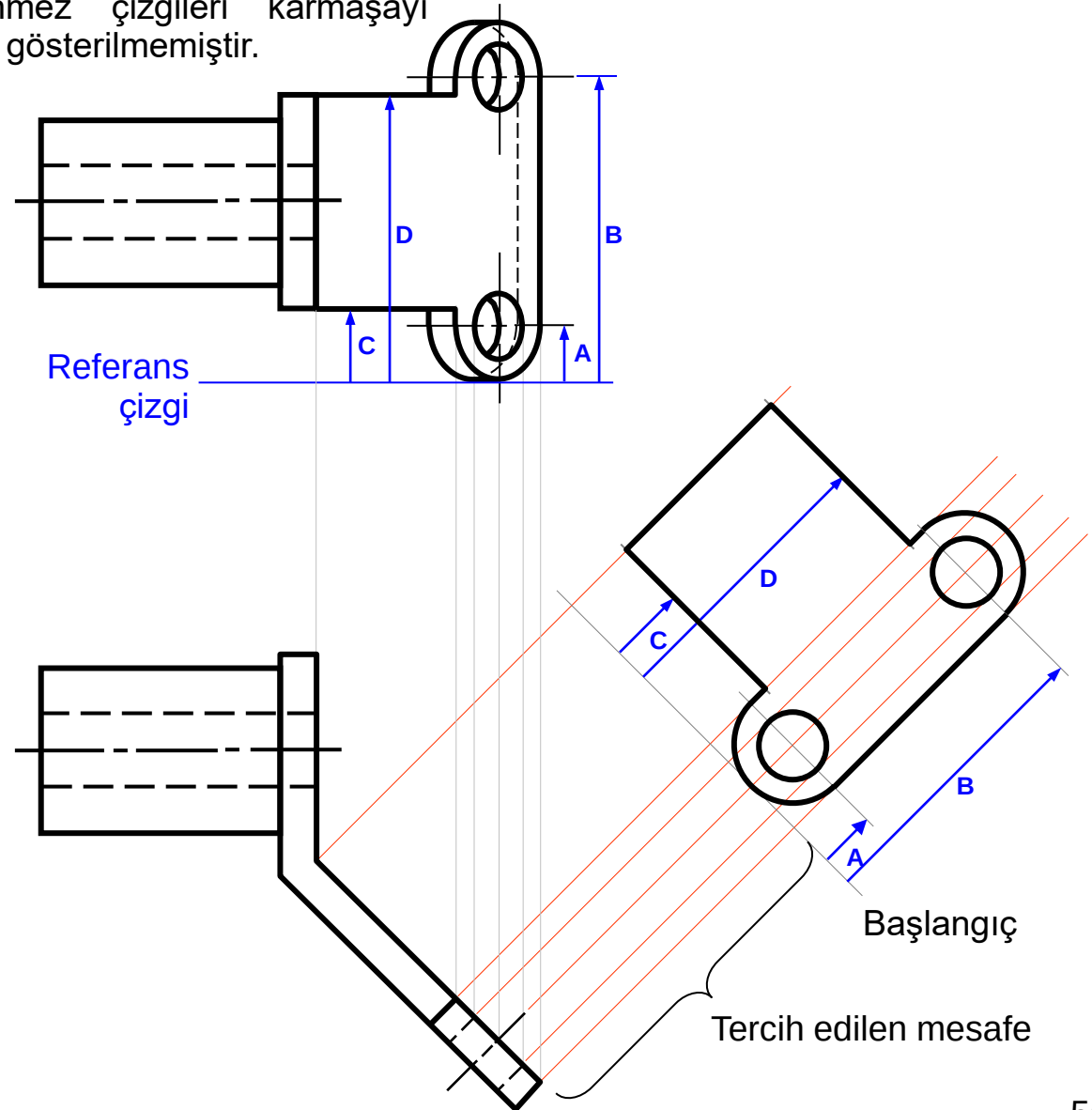
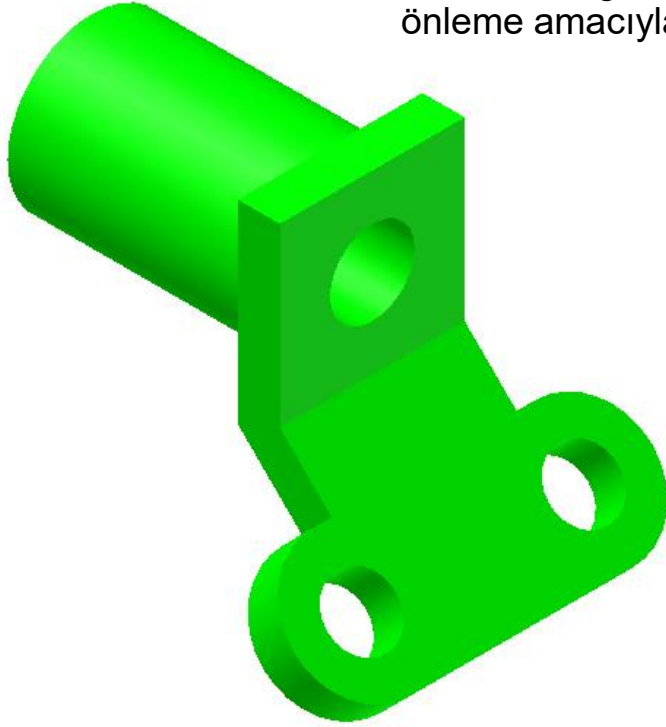
Örnek 1

■ Üst ve Yan görünüşlerden gerçek boyutu ölçülebilirmi ?



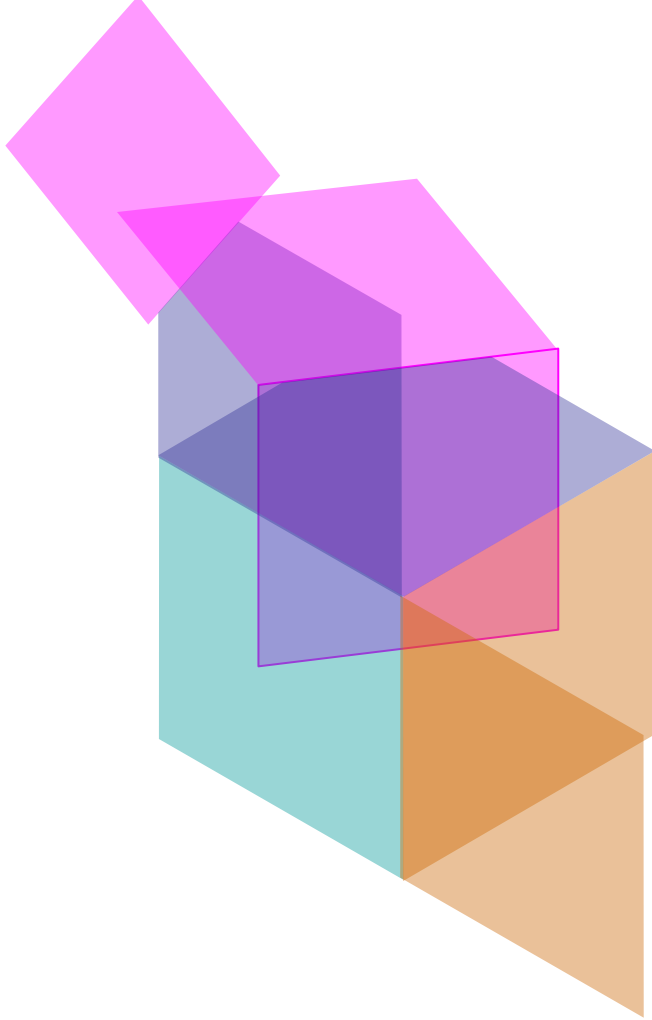
Örnek 2

Deliklerin görünmez çizgileri karmaşayı önleme amacıyla gösterilmemiştir.

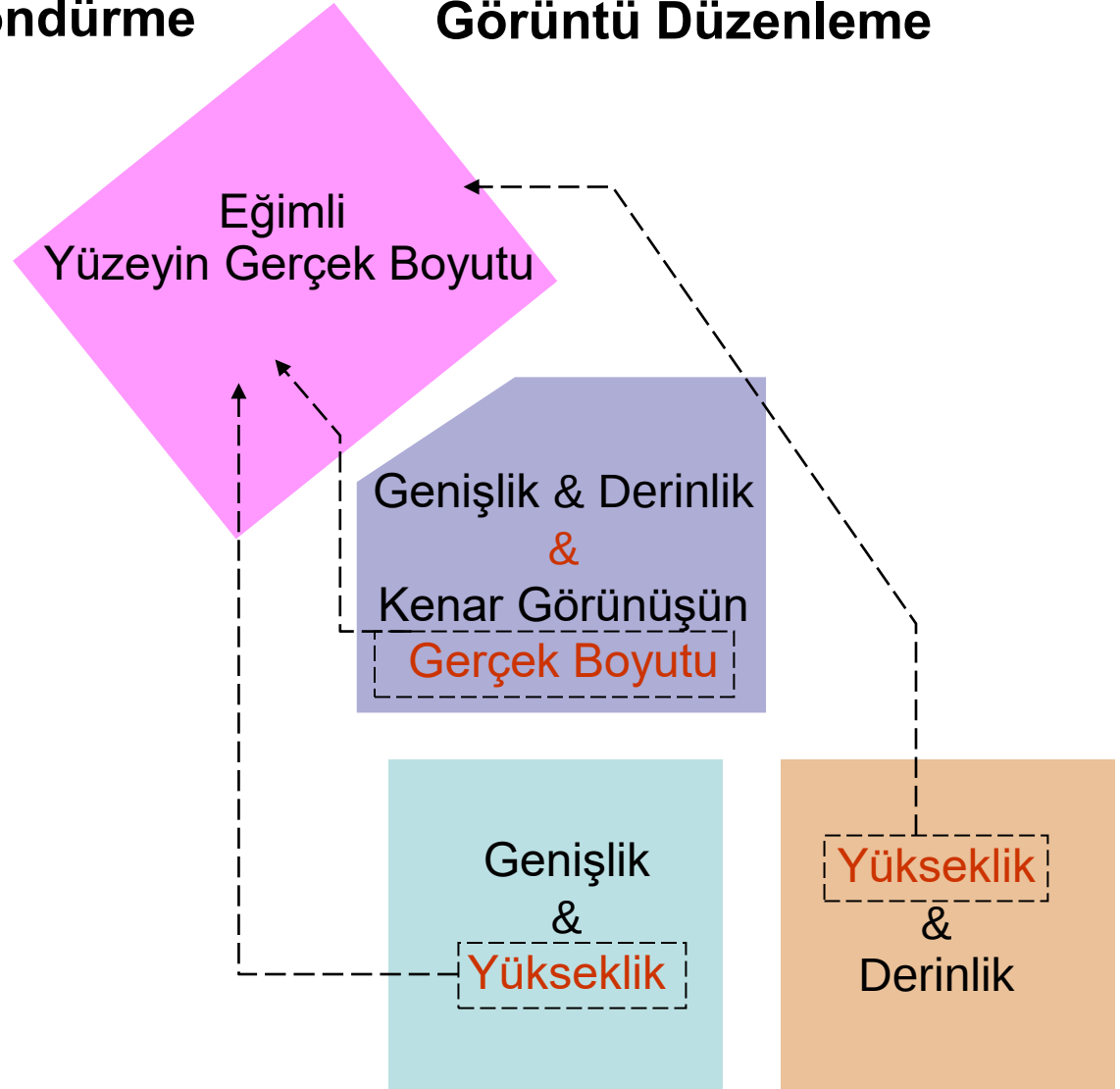


Üst Görünüşe Komşu Yardımcı Görünüşler

Cam Kutu modeli ve Döndürme



Görüntü Düzenleme

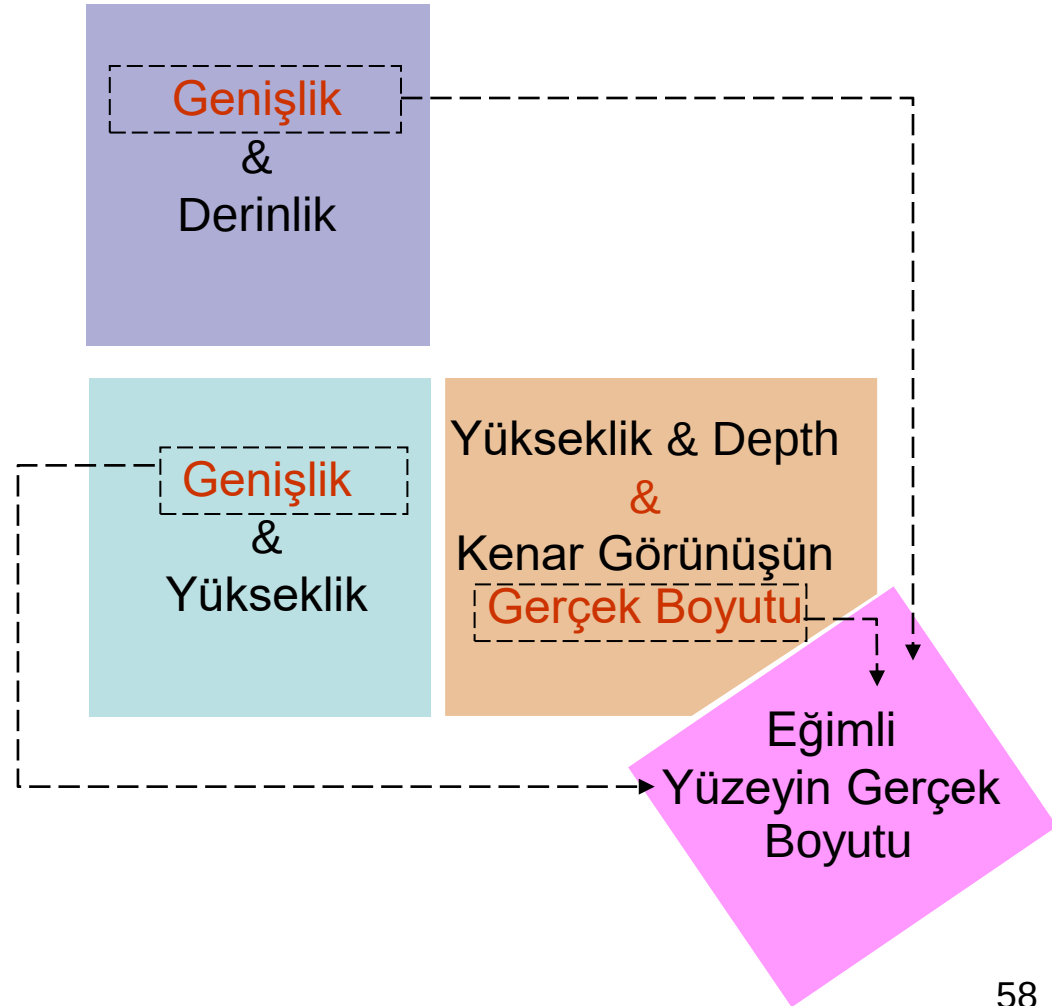


Yan Görünüşe Komşu Yardımcı Görünüşler

Cam Kutu modeli ve Döndürme



Görüntü Düzenleme



Örnek

